# AN INTEGRATED AI AND IOT-BASED SMART MALDIVE FISH PROCESSING SYSTEM FOR QUALITY ASSESSMENT AND PROCESS OPTIMISATION

**INDIVIDUAL THESIS**

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# DECLARATION AND SUPERVISOR CERTIFICATION

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# ABSTRACT

Maldive fish is a cured, hard-dried skipjack tuna product that is central to Sri Lankan cuisine. Its quality critically depends on tightly coupled processing stages, among which boiling and brine-salting determine muscle structure, energy consumption, and most crucially, the distribution of salt into the deep tissue that later protects the product during drying. In conventional cottage facilities, this stage is controlled by subjective operator judgement, which leads to inconsistent boiling durations, uneven salt penetration, avoidable fuel use, and flaking losses.

This thesis presents the design, implementation, and evaluation of an Intelligent Boiling and Brine-Salting Process Optimization module as an integral tier of the Aqua Sense Smart Maldive Fish Processing System. The module takes input parameters such as batch mass, the mean thickness of the fish cuts and the freshness grade of the produce (generated by the AI computer vision model of the raw-intake tier) to compute the dynamic boiling duration using an empirical power-law regression model with a freshness-dependent thermal offset (0 min for Very Fresh and +3.5 min for Fresh), and the needed sodium chloride ("NaCl") mass to hold a brine concentration of 12%–15% (adaptable ranges from 3% to 20% based on target requirements) and supervise core temperature between 92 °C and 96 °C with digital PT100 probes and ESP32 micro-controllers. The batch record and telemetry data are published via MQTT protocols to a central dashboard in support of complete traceability.

The module was evaluated on production batches (with trials in vertical positioning methods in cylindrical vessels to optimize blood drainage, heat distribution, and moisture diffusion) at a commercial facility in southern Sri Lanka and compared to conventional manual batches. Mean boiling time reduced from 58.4 ± 6.2 min to 42.1 ± 1.8 min (−27.9%), energy consumption reduced from 3.82 to 2.94 MJ/kg (−23.0%), the coefficient of variation of deep-tissue salt penetration reduced from 18.6% to 5.2%, core muscle firmness increased from 3.4 ± 0.6 to 4.6 ± 0.2 on a 1–5 scale, and flaking and over-boil loss reduced from 6.8% to 1.1%. The results indicate that a transparent, affordable, empiric controller is capable of accelerating, optimizing, uniforming, and enabling traceability of a traditional processing stage.

**Keywords:** Maldive fish, AquaSense, boiling optimisation, brine-salting, salt diffusion, Random Forest, PT100, Internet of Things, process control

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I sincerely thank the owners and staff of the commercial processing facility in southern Sri Lanka for allowing the production trials to be carried out and for sharing their practical knowledge and experience of Maldive fish processing.

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# TABLE OF CONTENTS

**1 INTRODUCTION** ..... 1
1.1 Overall project context ..... 2
1.2 Individual component context ..... 3
1.3 Problem statement ..... 4
1.4 Research gap ..... 5
1.5 Aim and objectives ..... 6
1.6 Research questions or hypotheses ..... 7
1.7 Scope limitations and assumptions ..... 8
1.8 Individual contribution ..... 9
1.9 Thesis structure ..... 10
**2 LITERATURE REVIEW AND RELATED WORK** ..... 12
2.1 Review method and scope ..... 12
2.2 Domain and theoretical foundations ..... 12
2.3 Related research ..... 16
2.4 Existing systems, products and techniques ..... 18
2.5 Gap synthesis and conceptual framework ..... 19
**3 RESEARCH METHODOLOGY** ..... 20
3.1 Research design ..... 20
3.2 Data, participants or study materials ..... 21
3.3 Experimental or study procedure ..... 22
3.4 Measures and evaluation metrics ..... 24
3.5 Data analysis ..... 24
3.6 Ethics, privacy, security and responsible AI ..... 25
3.7 Reproducibility and research artefacts ..... 25
**4 PRODUCT AND SOLUTION DESIGN** ..... 26
4.1 Requirements ..... 26
4.2 Component architecture within the complete solution ..... 26
4.3 Design alternatives and decisions ..... 27
4.4 Detailed design ..... 27
4.5 Standards and professional practice ..... 34
**5 IMPLEMENTATION, INTEGRATION AND TESTING** ..... 35
5.1 Development environment and technology stack ..... 35
5.2 Component implementation ..... 35

5.3 Integration with the other components ..... 37
5.4 Verification and software testing ..... 38
5.5 Deployment and operation ..... 40
**6 RESULTS AND VALIDATION** ..... 42
6.1 Research results ..... 42
6.2 Product and technical results ..... 44
6.3 User or stakeholder validation ..... 46
6.4 Integration results ..... 46
6.5 Summary of findings ..... 47
**7 DISCUSSION** ..... 48
7.1 Interpretation of findings ..... 48
7.2 Comparison with literature baselines and existing solutions ..... 49
7.3 Research contribution ..... 50
7.4 Product contribution ..... 50
7.5 Limitations and threats to validity ..... 51
7.6 Implications ..... 51
**8 COMMERCIALISATION AND SUSTAINABILITY** ..... 53
8.1 Value proposition and target users ..... 53
8.2 Market and competitive position ..... 53
8.3 Business or sustainability model ..... 54
8.4 Scalability, intellectual property and compliance ..... 55
8.5 Contribution to the Sustainable Development Goals ..... 56
**9 CONCLUSION AND FUTURE DIRECTIONS** ..... 57
9.1 Conclusion ..... 57
9.2 Achievement of objectives ..... 57
9.3 Individual research and product contributions ..... 57
9.4 Future directions ..... 58
**REFERENCES** ..... 60
**GLOSSARY** ..... 62
**APPENDICES** ..... 63
Appendix A Research instruments or extended results ..... 63
Appendix A.1 Operator feedback questionnaire (proposed instrument) ..... 64
Appendix B Academic integrity and AI use disclosure ..... 65

# LIST OF FIGURES

Figure 1.1: Four-tier processing pipeline of the Smart Maldive Fish Processing System, with the component reported in this thesis highlighted. ..... 3

Figure 1.2: Position of the boiling and salting module and its data exchanges with the neighbouring tiers. ..... 4

Figure 2.1: Conceptual and component architecture of the boiling and salting module. .....19

Figure 3.1: Research process followed in this study. .....20

Figure 4.1: Predicted boiling duration versus mean thickness for several batch masses ($\delta = 0$). .....28

Figure 4.2: Predicted boiling duration versus batch mass for several thicknesses ($\delta = 0$). ....28

Figure 4.3: Salt mass required for a given water mass within the 12%–15% brine window. .....30

Figure 4.4: Control flow of the boiling and salting module. .....31

Figure 6.1: Comparison of conventional and optimised processing for the principal metrics. .....42

# LIST OF TABLES

Table 1.1: Comparison of approaches to the boiling and salting stage. ..... 6
Table 2.1: Summary of the literature and its relevance to this component. ..... 18
Table 2.2: Existing approaches relevant to boiling and salting of fish. ..... 18
Table 3.1: Equipment and instrumentation used in the boiling and salting experiments. ..... 21
Table 3.2: Evaluation metrics for the boiling and salting module. ..... 24
Table 4.1: Functional requirements. ..... 26
Table 4.2: Non-functional requirements. ..... 26
Table 4.3: Design alternatives considered for the boiling-duration model. ..... 27
Table 4.4: Boiling duration $t_b$ (min) for Very Fresh batches ($\delta = 0$). ..... 29
Table 4.5: Batch record schema. ..... 31
Table 4.6: Worked examples of parameter computation. ..... 32
Table 4.7: Failure mode and effects summary. ..... 32
Table 5.1: Technology stack of the boiling and salting module. ..... 35
Table 5.2: Interfaces between the boiling and salting module and other components. ..... 38
Table 5.3: Model verification test cases (expected values computed from Equation 4.1). ..... 38
Table 5.4: Functional and interface tests. ..... 39
Table 5.5: End-to-end integration scenarios. ..... 39
Table 6.1: Experimental comparison: conventional manual versus proposed optimised processing. ..... 42
Table 6.2: Derived relative improvements. ..... 44
Table 6.3: Effect of input errors on predicted boiling time (reference 40.0 min). ..... 44
Table 6.4: Per-batch effect for a 15 kg batch (derived from Table 6.1). ..... 45
Table 6.5: System-level results reported in the group paper [17]. ..... 46
Table 6.6: Summary of how the results relate to objectives and hypotheses. ..... 47
Table 7.1: Positioning of the proposed module relative to existing approaches. ..... 50
Table 7.2: Threats to validity and mitigations. ..... 51
Table 8.1: Value to stakeholders. ..... 53
Table 8.2: Competitive comparison of alternatives available to a processor. ..... 54
Table 8.3: Illustrative payback framework (values to be supplied by the facility). ..... 54
Table 8.4: Commercial risks and mitigations. ..... 55
Table 8.5: Contribution to the United Nations Sustainable Development Goals [19]. ..... 56
Table 9.1: Achievement of the sub-objectives. ..... 57
Table 9.2: Glossary of terms. ..... 62
Table A.1: Model constants and parameters. ..... 63

Table A.2: Extended boiling-duration table, Very Fresh (δ = 0), minutes. Add 3.5 min for Fresh. ....63
Table A.3: Salt mass (kg) for common water masses. ....63
Table A.4: Batch record template. ....64
Table A.5: Proposed MQTT message for the batch record (illustrative schema). ....64

# LIST OF ABBREVIATIONS

**Table B.1: Abbreviations.**

| Abbreviation | Meaning |
|---|---|
| **AI** | Artificial Intelligence |
| **ADC** | Analog-to-Digital Converter |
| **CNN** | Convolutional Neural Network |
| **CV** | Coefficient of Variation |
| **DHT22** | Digital humidity and temperature sensor |
| **ESP32** | Espressif 32-bit microcontroller |
| **IoT** | Internet of Things |
| **MCU** | Microcontroller Unit |
| **MQ-135** | Metal-oxide gas sensor |
| **MQTT** | Message Queuing Telemetry Transport |
| **NaCl** | Sodium chloride |
| **NIR** | Near-infrared |
| **PT100** | Platinum resistance thermometer (100 Ω) |
| **RH** | Relative humidity |
| **SDG** | Sustainable Development Goal |
| **TVB-N** | Total Volatile Basic Nitrogen |
| **VOC** | Volatile Organic Compound |
| $\mathbf{a_w}$ | Water activity |

# 1 INTRODUCTION

Maldive fish, locally known as **Umbalakada**, is a cured and hard-dried product made from skipjack tuna (*Katsuwonus pelamis*). It is a foundational ingredient in Sri Lankan cuisine and an important cottage-industry product, particularly in coastal communities. Converting perishable raw tuna into a shelf-stable product requires a sequence of tightly coupled stages: sorting and grading of the raw fish, evisceration and cutting, boiling in brine, salting, dehydration, and dry storage [1]. The quality of the finished product is the cumulative outcome of all of these stages, and an error introduced early in the chain is rarely corrected later.

Among these stages, **boiling and brine salting** occupies a pivotal position. Boiling coagulates the muscle proteins, inactivates the endogenous autolytic enzymes that would otherwise continue to degrade the flesh, and begins the migration of sodium chloride (NaCl) into the deep muscle tissue. The salt that enters the tissue during this stage lowers the water activity ($a_w$) of the fish and is the principal barrier to microbial growth during the several days of drying that follow. If boiling is too short, the core of the fish is under-cooked, salt penetration is uneven, and the product dries slowly. If boiling is too long, the muscle fibres rupture, flaking and yield loss increase, and fuel is wasted.

In conventional cottage facilities this stage is controlled almost entirely by operator experience. Boiling duration is judged by eye, salt is applied as an unmeasured quantity or in brines of variable strength, and the water volume is rarely standardised against the batch mass. The result is substantial batch-to-batch variability in boiling time, energy consumption, tissue texture and salt distribution. The research summarised in this thesis addresses precisely this stage, by developing and evaluating an **empirical, data-driven control module for boiling duration and brine salting**.

The work was carried out as one individual component of a larger group project, the **AI and IoT-Based Smart Maldive Fish Processing System** [17], in which four students each developed one tier of an end-to-end digital pipeline. This thesis reports the design, implementation and evaluation of the second tier of that pipeline: the **Intelligent Boiling and Salt Distribution Optimisation** module. The module accepts

the batch mass, the mean thickness of the fish cuts and the freshness grade produced by the raw-intake tier, and returns the boiling duration, the salt mass and the brine concentration that should be used for that batch, while supervising the boiling temperature through digital PT100 probes.

The remainder of this chapter positions the component within the overall project, states the research problem and gap, defines the aim, objectives and research questions, sets out the scope, limitations and assumptions, and summarises the individual contribution of the author.

### 1.1 Overall project context

The Maldive fish industry in Sri Lanka remains largely artisanal. Fish are sorted, boiled, salted, and dried in cottage facilities, where each stage depends on the experience of individual operators. Raw freshness is judged from the eyes, gills, and skin; boiling time is estimated by observation; salt is applied by handfuls or in brines whose strength is rarely measured; and drying takes place in open yards exposed to monsoonal rain and humidity [2]. Processing losses exceeding 20% have been reported for such unmonitored operations, and finished batches are graded by subjective organoleptic evaluation.

The group project proposes an integrated cyber-physical architecture that couples Artificial Intelligence (AI) and the Internet of Things (IoT) across the complete production lifecycle. As illustrated in Figure 1.1, the system consists of four interconnected tiers that map directly onto the industrial workflow:

1. multimodal raw tuna inspection utilizing an **AI computer vision model** for freshness grading;
2. intelligent boiling and brine-salting process optimization;
3. an **ESP32-driven IoT edge network** monitoring the boiling and drying microclimate parameters (using PT100 temperature sensors and precise salinity controls); and
4. finished-product quality assessment combined with cloud-based dashboard traceability and shelf-life monitoring.

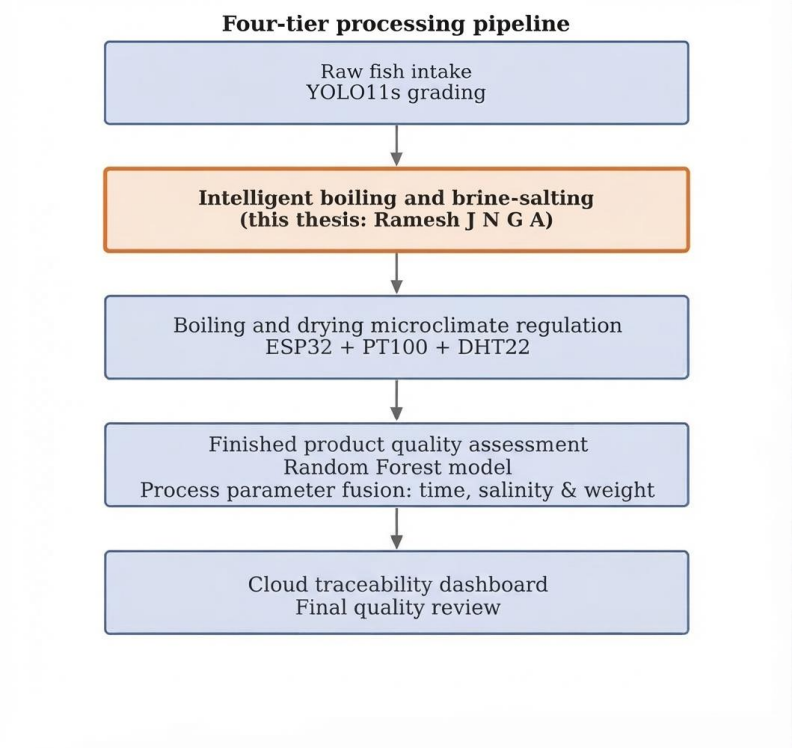



**Figure 1.1: Four-tier processing pipeline of the Smart Maldive Fish Processing System, with the component reported in this thesis highlighted.**

Across 20 production batches of 15 kg raw tuna each, evaluated over four months at a commercial processing facility in southern Sri Lanka, the integrated platform achieved an overall raw grading mAP@0.5 of 95.8%, a 27.9% reduction in boiling time (with a 23.0% reduction in thermal energy), a reduction in deep-tissue salt variation from 18.6% to 5.2%, and a final product grading accuracy of 94.0% [17]. The component reported here is responsible for the second and third of these quantitative outcomes.

## 1.2 Individual component context

The individual component of this thesis is the **Intelligent Boiling and Brine-Salting Process Optimisation** module. It sits between the raw-intake tier, which classifies each incoming batch as Very Fresh, Fresh or Spoiled, and the drying tier, which monitors the dehydration microclimate. Its position in the pipeline is shown in Figure 1.2.

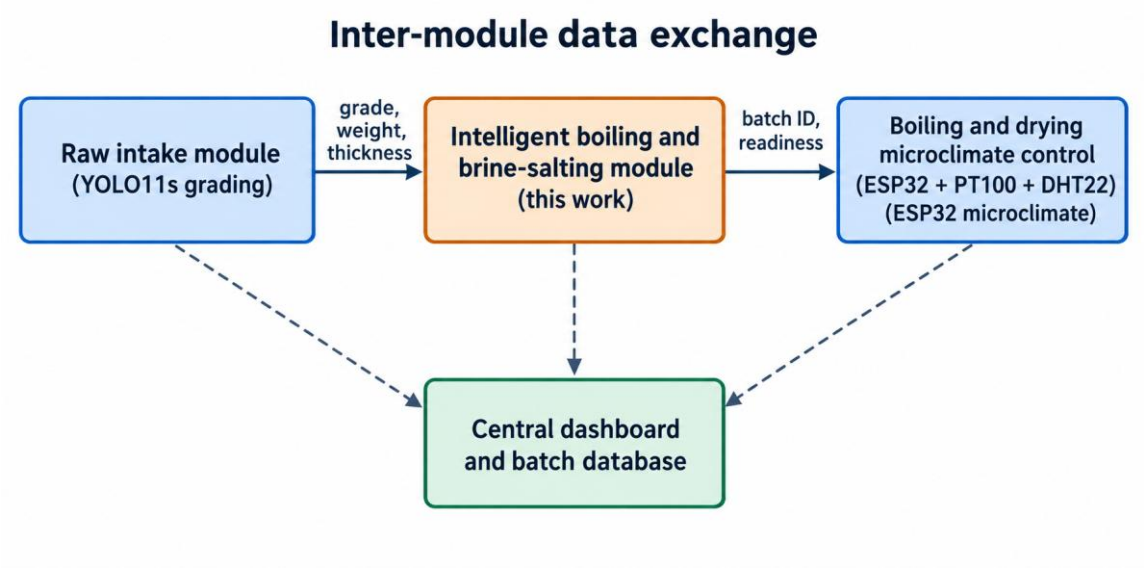



**Figure 1.2: Position of the boiling and salting module and its data exchanges with the neighbouring tiers.**

The module takes three principal inputs: the **batch mass** $M_{batch}$ (kg), the **mean thickness** $d_m$ (cm) of the fish cuts, and the **freshness grade** produced upstream. From these it computes the boiling duration $t_b$ (minutes) from an empirical regression, and the salt mass $M_s$ (kg) required to maintain a brine concentration of 12%–15% salt-to-water weight ratio. The deep-tissue salt target is NaCl ≥ 12%. During boiling, core temperature is logged by digital PT100 probes to ensure that thermal equilibrium is held between 92 °C and 96 °C.

The module outputs a complete set of process parameters and a batch record, which is forwarded to the central dashboard. In this way the stage that was previously controlled by memory and habit becomes a parameterised, logged and repeatable operation, which is also a precondition for the traceability objective of the wider project.

### 1.3 Problem statement

The step involving boiling and salting in the processing of Maldive fish relies on estimates given by the operator, and this in turn gives rise to three connected problems. First, the thermal treatment is not uniform since the boiling times vary considerably from one operator to another and from one batch to another. In the conventional batches which were used as the standard in this study, the average boiling time was 58.4 ± 6.2 minutes. If the boiling continues for too long, the muscles break down and

this leads to loss by flaking, while if the boiling is not long enough the centre of the fish remains undercooked and this in turn hinders the following dehydration. Second, the distribution of salt is uneven; if the dry salting is not carefully measured or if the brines differ, sodium chloride will migrate in an uneven manner into the deeper tissues. The coefficient of variation for salt penetration in the baseline case was 18.6%, meaning that in certain areas of the product the reduced water activity offers inadequate protection while in other areas it causes over-salting. Third, there is unnecessary energy use and a loss of yield. Under the baseline conditions the thermal energy consumption was 3.82 MJ per kg of fish and the baseline loss due to flaking and over-boiling was 6.8 per cent; these figures are not monitored in conventional facilities and so the waste goes unnoticed by the operator. It is therefore essential to have a control method which is quantitative, inexpensive, and verifiable since it should convert the properties of the batch received into the appropriate boiling and salting conditions and at the same time monitor the process by means of measured temperature; the method must be simple enough to be used in a cottage environment but at the same time rigorous enough to be validated against traditional methods..

### 1.4 Research gap

Previous studies have looked at the use of computer vision for classifying marine freshness [3], [7], electronic-nose arrays for detecting fish spoilage [9], hyperspectral methods for assessing freshness [10], and automated drying cabinets. These various approaches have still remained separate [17]. The literature regarding machine vision in the context of food quality evaluation [1], [2] focuses on measuring quality rather than on controlling the process parameters which determine it, and the research that deals with the thermal processing of fish generally involves raw or chilled fish, canning, or the industrial cooking of other types of fish.

Very little or no published research deals with the parametric control of boiling and brine salting for Maldive fish, since three features set this problem apart from other fish-cooking processes. Firstly, the product is not meant to be eaten right after cooking; boiling is an intermediate stage whose function is to prepare the fish for the following dehydration. Secondly, deep-tissue salt penetration, not just doneness, is a key result. Thirdly, the raw

material varies in both size and freshness, necessitating that the process parameters be adjusted for each batch rather than remaining fixed.

**Table 1.1: Comparison of approaches to the boiling and salting stage.**

| Aspect | Manual practice | Fixed-recipe cooking | Proposed adaptive module |
|---|---|---|---|
| Basis of boiling time | Operator judgement | Single fixed time | Regression on thickness, mass and freshness |
| Salt dosing | Unmeasured / variable brine | Fixed salt quantity | Brine target of 12%–15% with computed salt mass |
| Temperature supervision | Visual / none | Thermostat set-point | PT100 core probe, 92–96 °C window |
| Adaptation to freshness | None | None | Thermal offset by grade |
| Energy awareness | None | Indirect | Energy logged per batch (MJ/kg) |
| Traceability | Absent | Limited | Digital batch record |

The research gap addressed by this thesis is therefore the absence of an empirically derived, batch-adaptive boiling and salting control method for Maldive fish that is evaluated on full-scale production batches and linked to upstream freshness grading and downstream drying monitoring.

## 1.5 Aim and objectives

### Main objective

The main objective of this research is to design, implement and evaluate an empirical process-optimisation module that computes batch-specific boiling duration and brine-salting parameters for Maldive fish, so as to minimise structural muscle degradation and thermal energy use while achieving uniform deep-tissue salt penetration (NaCl ≥ 12%).

### Sub-objectives

1. To characterise the boiling and brine-salting stage and to identify the process variables (boiling duration, brine concentration, core temperature, batch mass, cut thickness and incoming freshness grade) that influence yield, texture, energy use and salt distribution.

2. To develop an empirical regression model for the boiling duration $t_b$ as a function of mean thickness, batch mass and freshness grade.
3. To define a brine-salting rule that maintains a 12%–15% salt-to-water weight ratio and computes the corresponding salt mass $M_s$.
4. To design a temperature-supervision mechanism based on digital PT100 probes that confirms thermal equilibrium between 92 °C and 96 °C.
5. To integrate the module with the raw-intake tier (freshness grade and batch weight) and with the central dashboard (batch record and traceability).
6. To evaluate the module on 20 full-scale batches against conventional manual batches using boiling time, energy consumption, salt-penetration variation, core muscle firmness and flaking loss.
7. To analyse the sensitivity, limitations and commercial potential of the module and to identify directions for future work.

## 1.6 Research questions or hypotheses

### Research questions

**RQ1:** Can boiling duration be predicted from mean cut thickness, batch mass and freshness grade with a simple empirical power-law model that is practical for cottage facilities?

**RQ2:** To what extent does model-based boiling duration reduce boiling time and thermal energy consumption compared with conventional manual practice?

**RQ3:** Does maintaining the brine concentration within 12%–15% salt-to-water weight ratio reduce the variation of salt penetration in deep muscle tissue?

**RQ4:** How does optimised processing affect structural quality, expressed as core muscle firmness and flaking loss?

**RQ5:** Can the module exchange data reliably with the raw-intake and drying tiers so that boiling parameters adapt to the incoming freshness grade?

### Research hypotheses

**H1:** Model-based boiling duration will be shorter and less variable than manually judged duration.

**H2:** Model-based boiling will reduce thermal energy consumption per kilogram of fish.

**H3:** Controlled brine concentration (12%–15%) will reduce the coefficient of variation of salt penetration in the deep core tissue.

**H4:** Reduced boiling over-extension and vat agitation will increase core muscle firmness and reduce flaking and over-boil loss.

### 1.7 Scope limitations and assumptions

#### Scope

The scope of the component includes the following activities:

1. Empirical modelling of boiling duration as a function of thickness, batch mass and freshness grade.
2. Computation of salt mass and brine concentration for a 12%–15% salt-to-water weight ratio.
3. Supervision of core temperature using digital PT100 probes with a target window of 92 °C to 96 °C.
4. Logging of batch parameters, temperature, energy consumption and outcome measures.
5. Interfacing with the raw-intake tier (grade and weight) and the central dashboard.
6. Experimental comparison with conventional manual batches over 20 production batches of 15 kg each.

The scope **excludes** raw fish grading (tier 1), drying microclimate control (tier 3), finished-product grading and shelf-life prediction (tier 4), and the development of automated vat hardware.

#### Limitations

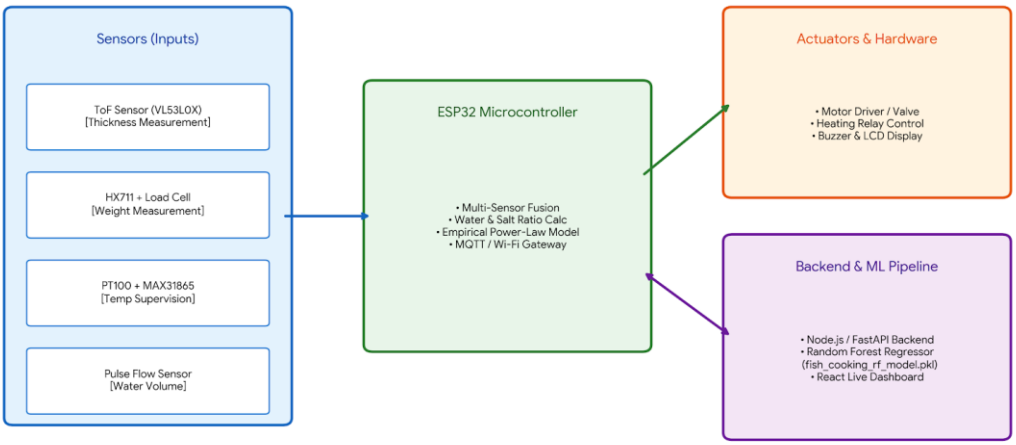
1. **Empirical nature of the model.** The regression coefficients ($\alpha$ = 40.0 min, exponents 1.4 and 0.3) were obtained empirically for skipjack tuna loins processed at the test facility; they have not been validated for other species, cut geometries or vat designs.

2. **Restricted freshness range.** A thermal offset is defined only for Very Fresh ($\delta = 0$) and Fresh ($\delta = +3.5$ min) grades. Spoiled batches are rejected upstream and are not boiled.
3. **Measurement of salt penetration.** Salt penetration is expressed as a coefficient of variation across deep-tissue samples; the sampling plan is limited by the number of samples that can be analysed per batch.
4. **Single facility.** All trials were carried out at one commercial facility in southern Sri Lanka over four months, so seasonal and regional variation in raw material is only partly represented.
5. **Sensory scoring.** Core muscle firmness is scored on a 1–5 scale, which retains an element of subjectivity even when scored by trained assessors.

### Assumptions

1. Batch mass is available from the raw-intake weighing step and the mean thickness can be measured or estimated for each batch.
2. Brine concentration expressed as a salt-to-water weight ratio is a sufficient proxy for the driving force of salt diffusion.
3. PT100 probe readings represent the true core temperature of the fish within the stated tolerance.
4. Heat input to the vat can be adjusted by the operator or by an actuator when temperature leaves the 92–96 °C window.

## 1.8 Individual contribution




As an individual member of the project team, my main contribution was the development of the **Intelligent Boiling and Brine-Salting Optimisation** component. The major contributions are as follows.

1. **Requirement analysis.** Identification of the process variables and quality outcomes that matter in the boiling and salting stage and definition of measurable targets (time, energy, salt variation, firmness and loss).
2. **Empirical boiling model.** Formulation and calibration of the regression for boiling duration in Equation (2.1) of the group paper, reproduced as Equation (4.1) in this thesis, including the freshness-dependent thermal offset.
3. **Brine-salting rule.** Definition of the 12%–15% brine window and the procedure for converting it into a salt mass for a given water volume.
4. **Temperature supervision.** Integration of digital PT100 probes and definition of the 92–96 °C thermal-equilibrium window.
5. **Integration.** Specification of the data interfaces with the raw-intake tier, the drying tier and the dashboard.
6. **Experimental evaluation.** Planning and execution of the comparison between optimised and conventional batches and analysis of the results reported in Chapter 6.

7. **Documentation.** Preparation of this thesis and contribution to the group research paper.

### 1.9 Thesis structure

This thesis is organised into nine chapters followed by references, a glossary and appendices.

**Chapter 1 – Introduction** presents the project context, problem, research gap, aim and objectives, research questions, scope and individual contribution.

**Chapter 2 – Literature Review and Related Work** reviews the theoretical foundations of boiling, salting and water activity, and related research on AI, sensing and process control in fish processing.

**Chapter 3 – Research Methodology** describes the research design, study materials, experimental procedure, evaluation metrics, analysis, ethics and reproducibility.

**Chapter 4 – Product and Solution Design** specifies requirements, architecture, design alternatives and the detailed design of the model and controller.

**Chapter 5 – Implementation, Integration and Testing** describes the implementation, integration with other tiers, verification and deployment.

**Chapter 6 – Results and Validation** reports the experimental results.

**Chapter 7 – Discussion** interprets the findings, compares them with the literature and states limitations and implications.

**Chapter 8 – Commercialisation and Sustainability** examines value proposition, market position, business model and Sustainable Development Goal contributions.

**Chapter 9 – Conclusion and Future Directions** summarises the work and proposes future research.

# 2 LITERATURE REVIEW AND RELATED WORK

## 2.1 Review method and scope

The literature review was conducted to identify research, technologies and methods relevant to the boiling and brine-salting stage of Maldive fish processing and to the wider AI and IoT-based processing system of which it is part. The review covers five areas: (i) the processing chain and quality attributes of cured, dried fish; (ii) the physical and chemical principles of thermal treatment and salt diffusion in fish muscle; (iii) machine vision and sensing methods for fish quality [1]–[3], [7]–[10]; (iv) IoT-based monitoring and control of food processing; and (v) empirical and data-driven process optimisation.

Sources were drawn from review articles and peer-reviewed studies on computer vision for food quality, deep learning, object-detection architectures, fish freshness datasets and the kinetics of drying and quality degradation of marine products [1]–[16]. The references retained in this thesis are those that directly informed the design choices in Chapter 4 or the positioning of the work in Section 2.5. Because the boiling and salting stage of Maldive fish is rarely addressed explicitly, the review inevitably relies on adjacent literature, and the transfer of conclusions from other species and products is treated with caution throughout.

## 2.2 Domain and theoretical foundations

### 2.2.1 Maldive fish processing chain

Maldive fish is produced from skipjack tuna in five phases: sorting, evisceration, controlled brine boiling, dehydration (open sun or chamber) and dry storage [1]. In the cottage sector, tuna are cut into loins, boiled in brine, cooled, and then dried for several days. The final product is dense, dark mahogany in colour and rock hard, with moisture below approximately 15% and a water activity low enough to prevent microbial growth at ambient temperature.

Each phase affects the next. Raw material of poor freshness has already begun to degrade through early proteolysis and bacterial action, which can precede visible discolouration. Boiling sets the protein structure and salt profile; an uneven profile will later dry unevenly and favour case-hardening, internal moisture retention and

histamine formation in humid conditions [2]. Dehydration kinetics and storage stability therefore depend on decisions taken at the boiling vat, which is why this stage is a high-leverage point for process control.

### 2.2.2 Thermal denaturation and enzyme inactivation

Heating fish muscle causes the sequential denaturation of its proteins. Myosin, the dominant myofibrillar protein, begins to unfold at moderate temperatures, and the collagen of the connective tissue shrinks and eventually gelatinises at higher temperatures. Denaturation expels water, firms the flesh and, if prolonged, causes fibre separation (flaking) and loss of soluble solids into the cooking liquor. Boiling at 92–96 °C is sufficient to inactivate the endogenous proteases and lipases that cause autolysis and to reduce the surface microbial load, which is the reason that boiling is described in this thesis as inhibiting autolytic enzymes and coagulating muscle proteins.

The time required to bring the **core** of the product to the target temperature depends on the thickness of the piece, because heat is transferred inwards by conduction. For a body of fixed shape, the characteristic conduction time grows approximately with the square of the characteristic length; in practice, irregular geometry, simultaneous shrinkage and the presence of bone and fat reduce the effective exponent. The empirical exponent of 1.4 used in this thesis for the thickness ratio is consistent with sub-quadratic scaling expected for such heterogeneous bodies.

### 2.2.3 Salt diffusion and water activity

The migration of sodium chloride into fish muscle is a diffusion-driven process, commonly described by Fick's second law [18]. In one dimension,

$$\frac{\partial \mathrm{C}}{\partial \mathrm{t}} = D_{\mathrm{s}} \frac{\partial^2 C}{\partial x^2} \tag{2.1}$$

where C is the salt concentration, t is time, x is distance into the tissue and $D_s$ is the effective diffusion coefficient. The effective coefficient is not a constant: it increases with temperature and depends on the state of the muscle proteins, and it is lower in tissue whose structure has been tightened by intense heat treatment. Consequently, both the boiling temperature and the duration influence how deep the salt travels, and an over-extended boil can in principle reduce the effective penetration rate even as it

wastes energy. The driving force is the concentration difference between brine and tissue, which explains why a controlled brine concentration, rather than unmeasured dry salt, is expected to give a more uniform profile.

Salt lowers water activity ($a_w$), defined as the ratio of the vapour pressure of water in the product to that of pure water. Microbial growth and many enzymatic reactions are strongly suppressed as $a_w$ falls. For Maldive fish, a deep-tissue NaCl content of at least 12% is used as the salt target in this research, and values of $a_w$ above about 0.75 are associated with under-dried, unstable product [17]. Because the later drying stage can only remove water, not correct a poor initial salt profile, the uniformity of salt distribution established during boiling is a decisive quality factor.

### 2.2.4 Empirical modelling of process time

When a full mechanistic model is impractical, empirical power-law regressions are widely used in food engineering to relate a process time to a small number of measurable properties. A typical form is $t = \alpha \cdot X_1^a \cdot X_2^b$, which becomes linear in logarithms and can be fitted by ordinary least squares. The advantage is transparency: operators and auditors can see how each variable influences the result, and the model can be recalibrated from routine production records. The limitation is that it is valid only within the range of the calibration data. The boiling model adopted here has exactly this form, with thickness and batch mass as the explanatory variables and an additive correction for freshness.

### 2.2.5 Sensing and IoT in process control

IoT architectures connect low-cost sensors and microcontrollers to a broker or database so that process data can be logged, visualised and acted on. In this project, an ESP32-based edge node already monitors the drying yard and publishes measurements via MQTT [17]. The boiling module follows the same philosophy: measure the physical state (core temperature), compute the setpoints (time, salt), and record everything. Digital PT100 platinum resistance probes are used because of their linearity, stability and accuracy in the 90–100 °C range.

### 2.2.6 Heat transfer into fish cuts and the thickness scaling

For a slab of thickness L heated from both faces, transient conduction is governed by the Fourier number, $Fo = \alpha_{th} \cdot t / (L/2)^2$, where $\alpha_{th}$ is the thermal diffusivity of the muscle. For a given target core-to-surface temperature ratio, the required Fo is approximately constant, so the time needed scales with $L^2$. Real fish cuts depart from this ideal in several ways: they are not slabs, but tapered loins with a rounded cross-section, for which the effective path length grows more slowly than thickness; their composition varies from lean muscle to fat-rich belly meat; and they are loaded together in a vat, so that neighbouring pieces shade each other from the circulating brine. The net effect is that an empirical exponent lower than two, such as the value of 1.4 adopted here, is physically plausible.

The mass exponent has a different origin. Increasing the batch mass at fixed vat volume increases the thermal load, so that the brine temperature recovers more slowly after the fish are loaded and the effective heating-up period is longer. Because this is a second-order effect compared with conduction inside each piece, a small exponent (0.3) is expected. The reference values of 4.5 cm and 15 kg are not physical constants; they simply normalise the ratios so that the constant $\alpha$ equals the boiling time at the reference condition.

### 2.2.7 Quality chemistry linked to boiling and salting

Several quality and safety attributes of the finished product are influenced by the boiling and salting stage. **Histamine formation** is a food-safety concern in scombroid fish such as tuna: it is produced by bacterial decarboxylation of free histidine, and is favoured by poor initial freshness, slow drying and high humidity [2]. A well-executed boil inactivates most vegetative bacteria and enzymes, and a sufficiently high deep-tissue salt content slows any subsequent growth. **Lipid oxidation** progresses during storage and is accelerated by heat and by prolonged exposure of fat-rich tissue to air; it determines the rancidity that limits shelf life, which is why the shelf-life model of the group project includes an activation energy for lipid oxidation [17]. **Texture** is governed by protein denaturation and moisture loss. Over-boiling produces a dry, fibrous texture and extensive flaking, whereas under-boiling leaves a soft core that dries slowly and may case-harden at the surface.

These interactions explain why the module's outcome measures were chosen as they were. Salt penetration uniformity is a proxy for microbiological stability, firmness and flaking loss are proxies for texture and yield, and the time and energy measures capture process efficiency. The measures cover the three dimensions of safety, quality and cost.

### 2.2.8 Process control concepts relevant to cottage-scale operations

Industrial process control distinguishes between **feed-forward** and **feedback** strategies. A feed-forward controller uses knowledge of the disturbance (here, batch mass, thickness and freshness) to choose the setpoints before the process starts. A feedback controller measures the output (here, core temperature) and corrects it. The module combines both: the regression is a feed-forward element, and the PT100 supervision is a feedback element. A third concept, **statistical process control**, uses recorded data to detect drift over time; the batch records produced by the module enable this at the level of monthly review.

## 2.3 Related research

### 2.3.1 Machine vision and deep learning in food quality

Reviews of computer vision for food quality evaluation show extensive success in sorting, grading and defect detection [1], [2], with deep learning improving feature extraction relative to hand-crafted descriptors [3]. Transfer-learned architectures such as ResNet, MobileNetV3 and EfficientNet [4], [13], [14] deliver high accuracy with modest computation, and single-stage object detectors [5], [6] allow real-time inspection on edge accelerators. These advances underpin tiers 1 and 4 of the group system, and they set the standard for the AI-based part of the pipeline into which this thesis' module is embedded.

### 2.3.2 Fish freshness assessment

Fish freshness has been assessed using colour histograms and classical classifiers such as k-NN and Naïve Bayes [7], using the DaFiF dataset combined with metal-oxide sensors [8], [9], using hyperspectral imaging with deep learning [10], and using deep CNNs [12]. All of these focus on raw or ice-stored fish. They are relevant to this thesis because the freshness grade they produce is the input used to select the thermal offset

δ in the boiling model. They do not, however, specify what should be done differently in the boiling vat for a fresher or less fresh batch, which is the question that this thesis addresses.

### 2.3.3 Quality of dried fish products

Digital image processing has been applied to the quality evaluation of dried fish [15], and kinetic modelling has been applied to moisture loss and quality degradation during drying of marine products [16]. These studies concern the downstream end of the chain. Their implicit message is that the properties of the dried product are strongly conditioned by earlier processing, and that deviations at the boiling stage propagate through the whole chain. They provide indirect support for treating boiling and salting as a controllable upstream determinant of final quality.

### 2.3.4 Process optimisation and control in food processing

Process optimisation in food engineering ranges from first-principles heat and mass transfer models through response surface methodology to modern machine-learning surrogates. In small and medium enterprises, the main barriers to adoption are data scarcity, the cost of instrumentation and the need for explainability. An empirical power-law model with a handful of parameters and a clearly bounded operating region is attractive under these constraints because it can be calibrated from a modest number of batches and is easily communicated to operators.

### 2.3.5 IoT architectures for small food processors

Published IoT architectures for food processing typically consist of a sensing layer (temperature, humidity, gas), a network layer (Wi-Fi, LoRa or cellular), a platform layer (broker, database) and an application layer (dashboard and alerts). In small facilities, the dominant constraints are cost, power, intermittent connectivity and operator skills. The group project adopted MQTT over 2.4 GHz 802.11b/g/n links for the drying node, which achieved a 99.6% packet delivery rate over a seven-day cycle [17]. The same messaging approach is used by this module, so that a single broker can aggregate records from all tiers.

A further lesson from the IoT literature is the importance of **edge autonomy**: the controller must keep working when the network is unavailable and synchronise

afterwards. This principle is reflected in NFR4 of Chapter 4 and in the local storage of batch records.

### 2.3.6 Summary of the literature

**Table 2.1: Summary of the literature and its relevance to this component.**

| Theme | Representative sources | Relevance to this thesis |
|---|---|---|
| Computer vision for food quality | [1], [2], [11], [15] | Context for upstream and downstream tiers |
| Deep learning and detectors | [3]–[6], [13], [14] | Technology of the freshness grade supplied to this module |
| Fish freshness datasets and sensing | [7]–[10], [12] | Source of the grade that sets the thermal offset |
| Kinetics of drying and degradation | [16] | Downstream consequence of salt and structure set during boiling |
| Diffusion theory | [18] | Basis for the brine-concentration rule |
| Integrated system | [17] | Overall architecture and experimental protocol |

## 2.4 Existing systems, products and techniques

**Table 2.2: Existing approaches relevant to boiling and salting of fish.**

| Approach | Main technique | Advantages | Main limitation |
|---|---|---|---|
| Manual practice | Operator judgement of boiling time and salt | No capital cost; knowledge-based | Subjective, variable, no records |
| Fixed-recipe industrial cooking | Fixed time/temperature programme | Repeatable | Ignores batch size and freshness variation |
| Thermostatic cookers | Temperature set-point control | Stable temperature | Does not choose duration or salt |
| Computer-vision inspection [1], [2] | Camera + classifier | Non-contact quality measurement | Measures, does not control |
| Gas/VOC sensing [9] | MOS sensor arrays | Detects spoilage chemistry | Not a process controller |
| Kinetic drying models [16] | Moisture and quality kinetics | Predicts drying behaviour | Downstream of boiling |
| Proposed module | Regression + brine rule + PT100 supervision | Batch-adaptive, simple, logged | Empirical; limited calibration range |

Thermostatic cookers and fixed-programme equipment can maintain a temperature, but none of the reviewed approaches computes the boiling duration and salt mass from the characteristics of the batch in front of the operator. The proposed module therefore complements rather than replaces thermostatic hardware: it provides the setpoints that such hardware lacks.

## 2.5 Gap synthesis and conceptual framework

The review shows that the technologies required for an intelligent boiling and salting module (empirical modelling, temperature sensing, IoT data logging and upstream freshness grading) all exist, but have not been combined for Maldive fish. Four specific gaps emerge: no published parametric model of boiling time for Maldive fish cuts; no brine-concentration rule linked to the deep-tissue salt target; no coupling between upstream freshness grading and downstream thermal treatment; and no evaluation of such a controller on full-scale production batches against conventional practice.

The conceptual framework of the component is:

**Raw batch (M, grade) + Cut geometry (d) → Parameter engine (t_{b}, M_{s}, brine 12–15%) → Boiling with PT100 supervision (92–96 °C) → Logged batch record → Drying tier**

The inputs are mass and thickness (physical), freshness grade (AI-derived) and temperature (sensed). The outputs are boiling duration, salt mass and an auditable record. The framework is deliberately conservative: it relies on a bounded empirical model and a measured safety constraint rather than on opaque prediction.

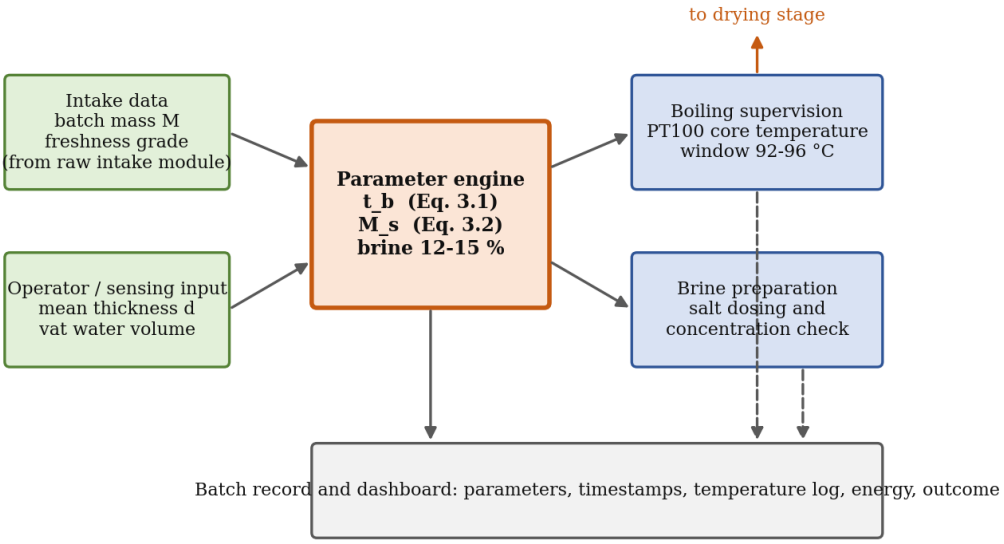



**Figure 2.1: Conceptual and component architecture of the boiling and salting module.**

# 3 RESEARCH METHODOLOGY

## 3.1 Research design

The research adopts a **design-and-development experimental approach**. A control module is first designed from the requirements of the boiling and salting stage, then implemented, and finally evaluated in full-scale production against conventional practice. The same facility, the same species and the same batch size were used for the optimised and the manual runs, so that the principal difference between the groups is the method used to decide the process parameters.

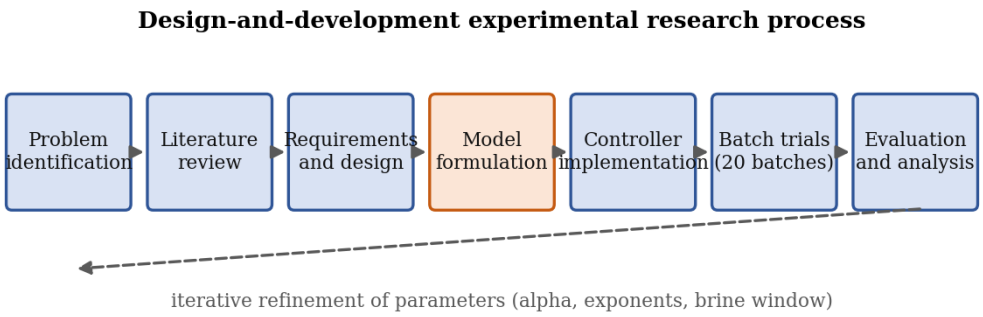



**Figure 3.1: Research process followed in this study.**

The process shown in Figure 3.1 began with the identification of the limitations of manual boiling and salting through observation at the facility. A literature review followed (Chapter 2), after which the requirements and architecture were defined (Chapter 4). The empirical model was then formulated, implemented as a controller, and used in 20 production batches over four months. Results were analysed against the conventional baseline (Chapter 6), and the model parameters were refined where the data indicated that the initial values were inadequate.

The design is a **comparative field experiment** rather than a laboratory study. This choice favours external validity, since the module is evaluated under real production conditions, at the cost of some experimental control: the raw material, ambient conditions and operators vary naturally between batches. This trade-off is discussed in Section 7.5.

### 3.2 Data, participants or study materials

The study involves no human participants in the sense of surveys or interventions on persons. The study materials are skipjack tuna, process equipment, sensors and the data generated by them.

#### 3.2.1 Raw material and batches

Twenty production batches were processed, each consisting of **15 kg of raw skipjack tuna** (*Katsuwonus pelamis*), over four months at a commercial processing facility in southern Sri Lanka. Each batch was assigned a freshness grade (Very Fresh or Fresh) by the raw-intake tier [17]. Batches graded Spoiled are rejected at intake and are therefore not boiled. The mean thickness of the cuts was measured for each batch with callipers on a representative sample of loins.

#### 3.2.2 Conventional comparison batches

The performance of the module was compared against **traditional manual facility batches** processed at the same facility, in which operators determined boiling duration, water volume and salt quantity by experience, without the module. The metrics for these batches form the Conventional Manual column of Table 6.1.

#### 3.2.3 Equipment and instrumentation

**Table 3.1: Equipment and instrumentation used in the boiling and salting experiments.**

| Item | Role | Key specification |
|---|---|---|
| Boiling vat | Brine boiling of fish batches | Capacity sufficient for 15 kg batches |
| Digital PT100 probes | Core and brine temperature logging | Target thermal window 92–96 °C |
| Platform scale | Batch mass $M_{batch}$ | Resolution to be recorded per scale [confirm model] |
| Callipers | Mean thickness $d_m$ | cm resolution |
| Salt and water measurement | Brine preparation | 12%–15% salt-to-water weight ratio |
| Energy measurement | Fuel/energy per batch | Reported in MJ/kg |
| Digital control unit | Parameter computation and logging | Runs the model of Chapter 4 |

#### 3.2.5 Batch eligibility and handling rules

To keep the experimental conditions comparable, simple eligibility and handling rules were applied. A batch was eligible if it consisted of whole skipjack tuna that had been eviscerated and cut into loins at the facility, if it had been graded Very Fresh or Fresh

by the raw-intake tier, and if its mass was close to the standard 15 kg. Spoiled batches were excluded by design because they are rejected before boiling. Batches were boiled on the same day as intake wherever possible to prevent additional deterioration between grading and boiling. After boiling, the batches were drained, cooled and passed to the drying tier under the same handling procedure as the conventional batches, so that differences in downstream handling would not confound the comparison.

Brine was prepared freshly for each experimental batch to the specified ratio. Where the facility's usual practice involved reusing brine, the reuse was recorded because it affects the effective concentration and therefore the salt penetration result.

### 3.3 Experimental or study procedure

Each optimised batch followed the same sequence of steps.

1. **Intake and grading.** The raw fish batch was weighed and graded by the raw-intake tier into Very Fresh, Fresh or Spoiled.
2. **Geometry measurement.** The mean thickness $d_m$ of the cuts was measured and entered with the batch mass.
3. **Parameter computation.** The control unit computed the boiling duration $t_b$ using Equation (4.1) and the salt mass $M_s$ for the brine target using Equation (4.2).
4. **Brine preparation.** Water was measured into the vat, salt was added to reach a salt-to-water weight ratio within 12%–15%, and the brine was heated.
5. **Boiling.** Fish were loaded, and core temperature was logged using digital PT100 probes to confirm thermal equilibrium between 92 °C and 96 °C. Mechanical agitation of the vat was reduced to limit muscle damage.
6. **Completion.** When the computed duration had elapsed, the batch was removed, cooled and handed to the drying tier. Energy consumption, boiling time and observations were recorded.
7. **Outcome measurement.** Salt penetration in deep tissue, core muscle firmness and flaking/over-boil loss were measured as described in Section 3.4.

Manual batches followed the facility's usual practice, with the same measurements taken after the fact.

### 3.3.1 Sampling plan and measurement procedure

For each batch, mean thickness was determined by measuring a representative sample of loins at their thickest section. Core temperature was monitored throughout boiling with PT100 probes inserted into the thickest part of selected pieces. After boiling and cooling, deep-tissue salt was determined on samples taken from the core of representative pieces, and the coefficient of variation across samples was computed for the batch. Core muscle firmness was scored on the 1–5 scale by trained assessors, using a fixed scoring sheet. Flaking and over-boil loss was determined from the mass of material separated from the pieces during and after boiling, expressed as a percentage of batch mass. [Add the exact number of loins/samples measured per batch if recorded.]

### 3.3.2 Calibration of the boiling-duration model

The boiling-duration equation is linear in logarithms after the freshness offset is subtracted. Taking natural logarithms of Equation (4.1):

$$\ln(t_{\mathrm{b}} - \delta) = \ln\alpha + a\ln\left(\frac{d_{\mathrm{m}}}{d_0}\right) + b\ln\left(\frac{M_{\mathrm{batch}}}{M_0}\right) \tag{3.1}$$

where a and b are the thickness and mass exponents (1.4 and 0.3 in the calibrated model). Given a set of batches for which a satisfactory boiling time has been established (for example, by the core temperature reaching the target and the firmness score being acceptable), ln α, a and b can be estimated by ordinary least squares. This provides a repeatable recalibration procedure for a new species, a new cut geometry or a different vat. Because thickness and batch mass are partly correlated, a design in which both vary independently across batches gives more reliable exponents than routine production data alone.

### 3.3.3 Instrument calibration and measurement uncertainty

PT100 probes were checked against a reference thermometer in boiling water before the trials, and the 92–96 °C window was applied to the verified readings. Scales were checked with a reference mass. Salt and water quantities were measured by mass to avoid volumetric error. Measurement uncertainty is not zero: a thickness measurement

error of ±0.5 cm at 4.5 cm, for instance, changes the predicted boiling time noticeably (Section 6.2), which is why the thickness measurement protocol is emphasised.

### 3.4 Measures and evaluation metrics

**Table 3.2: Evaluation metrics for the boiling and salting module.**

| Metric | Definition | Unit | Desired direction |
|---|---|---|---|
| Mean boiling time | Duration of boiling per batch (mean ± SD) | min | Lower, with lower SD |
| Energy consumption | Thermal energy per kg of fish | MJ/kg | Lower |
| Salt penetration coefficient of variation | Variation of salt content across deep-tissue samples | % | Lower |
| Core muscle firmness | Trained assessor score (mean ± SD) | 1–5 | Higher |
| Flaking / over-boil loss | Mass lost to flaking and over-boiling relative to batch mass | % | Lower |
| Temperature compliance | Share of boiling time with core temperature in 92–96 °C | % | Higher [report if logged] |

The **relative reduction** of a metric is computed as $(X_{manual} - X_{opt}) / X_{manual} \times 100\%$. The coefficient of variation is the standard deviation divided by the mean, expressed as a percentage.

### 3.5 Data analysis

Descriptive statistics (mean and standard deviation) were computed for each metric and group. Relative reductions were calculated as defined above. Because the comparison is between a conventional group and an optimised group processed in the same facility, the analysis concentrates on effect size and consistency (standard deviation and coefficient of variation) rather than on formal hypothesis tests alone. Where the number of batches per group allows, a two-sample test (Welch's t-test) is appropriate for the boiling time and firmness, and this is recommended in the further analysis proposed in Section 9.4.

The model itself was analysed by sensitivity analysis. Equation (4.1) was evaluated over the practical ranges of thickness (3.0–6.0 cm) and batch mass (5–30 kg) to determine how strongly each variable drives the predicted boiling time (Section 4.4). This analysis is deterministic and does not depend on experimental noise.

### 3.6 Ethics, privacy, security and responsible AI

The research does not involve human subjects, personal data or animal experimentation beyond the normal commercial slaughter of fish for food. Sensory scoring of firmness was carried out by facility staff as part of normal quality control. The facility consented to the collection and use of anonymised process data, and the dashboard stores no personal information.

Responsible-AI considerations are modest for this component because the model is simple and transparent. The inputs and parameters are visible to the operator, the model has a bounded operating range, and a **safety interlock** keeps boiling inside a measured temperature window regardless of the computed time. Food safety is protected by the rule that Spoiled batches are never boiled and by the thermal offset that adds 3.5 min for Fresh batches to ensure deep core sterilisation. Security of the data link between modules follows the MQTT practices of the wider project [17]. Operators remain able to override the controller, and any override is recorded.

### 3.7 Reproducibility and research artefacts

To make the work reproducible, the following artefacts are defined: the model equations and constants (Chapter 4 and Appendix A); the batch record schema (Appendix A); the experimental protocol (Section 3.3); and the result summary (Chapter 6). A reader with access to thickness, mass and grade for a batch can recompute the boiling duration and salt mass exactly. The sensitivity tables in this thesis were generated directly from Equation (4.1). Raw per-batch measurements are held by the project group and are available on request, subject to facility approval.

# 4 PRODUCT AND SOLUTION DESIGN

## 4.1 Requirements

Requirements were derived from observation of the cottage facility, from the process objectives in Chapter 1 and from the interface needs of the other tiers. They are summarised in Table 4.1 (functional) and Table 4.2 (non-functional).

**Table 4.1: Functional requirements.**

| ID | Requirement | Priority |
|---|---|---|
| FR1 | Accept batch mass, mean thickness and freshness grade as inputs | Must |
| FR2 | Compute boiling duration $t_b$ from thickness, mass and grade | Must |
| FR3 | Compute salt mass $M_s$ for a brine of 12%–15% salt-to-water weight ratio | Must |
| FR4 | Reject Spoiled batches before boiling | Must |
| FR5 | Log core temperature with digital PT100 probes and confirm the 92–96 °C window | Must |
| FR6 | Alert the operator when temperature leaves the window | Should |
| FR7 | Record batch parameters, temperatures, energy and outcome | Must |
| FR8 | Publish the batch record to the central dashboard | Should |
| FR9 | Allow operator override, with the override logged | Should |

**Table 4.2: Non-functional requirements.**

| ID | Requirement | Rationale |
|---|---|---|
| NFR1 | Computation must complete in under one second | Operator wait time |
| NFR2 | Model must be explainable to non-specialist operators | Adoption and trust |
| NFR3 | Low-cost, off-the-shelf sensors and controller | Cottage-industry affordability |
| NFR4 | Must function if the network link is temporarily lost | Facility connectivity |
| NFR5 | Parameters must be recalibratable from routine records | Maintainability |
| NFR6 | Operating range must be explicit; out-of-range inputs flagged | Safety and validity |

## 4.2 Component architecture within the complete solution

The module is organised as a pipeline with three layers: the **input layer** (batch mass and grade from the intake tier, thickness from measurement), the **parameter engine** (the model and brine rule), and the **supervision and recording layer** (PT100 monitoring, alerts and batch record). The architecture is shown in Figure 2.1 and its position in the overall pipeline in Figure 1.2. The module communicates with the raw-intake tier to receive grade and weight, and with the dashboard to publish the batch record. Its output to the drying tier is the batch identifier and a readiness signal.

The data flow for one batch is: intake record → parameter engine → operator display of setpoints → boiling with temperature logging → batch record → dashboard and drying tier. A local copy of the record is kept so that data are not lost if the network is unavailable (NFR4).

## 4.3 Design alternatives and decisions

**Table 4.3: Design alternatives considered for the boiling-duration model.**

| Alternative | Strengths | Weaknesses | Decision |
|---|---|---|---|
| Fixed time | Simplest | Ignores size and freshness | Rejected |
| Mechanistic heat-transfer simulation | Physically rigorous | Needs thermal properties and geometry; heavy to maintain | Rejected for cottage scale |
| Black-box machine-learning regressor | Flexible | Needs large data; low explainability | Rejected (data scarcity) |
| Empirical power-law regression with freshness offset | Few parameters; explainable; recalibratable | Valid only within calibrated range | **Selected** |

The empirical power law was selected because it satisfies NFR2 and NFR5 while being sufficiently accurate in the operating range. Two further decisions concern the brine and the temperature. The brine is specified as a **salt-to-water weight ratio of 12%–15%** because a ratio is simple to prepare with a scale and independent of vat size. The temperature is supervised by **digital PT100 probes** rather than thermocouples because of their stability and accuracy near 100 °C.

## 4.4 Detailed design

### 4.4.1 Boiling-duration model

The optimal boiling duration $t_b$ (minutes) is computed from the batch mass $M_{batch}$ (kg), the mean thickness $d_m$ (cm) and the intake freshness classification:

$$t_b = \alpha \cdot \left(\frac{d_m}{d_0}\right)^{1.4} \cdot \left(\frac{M_{batch}}{M_0}\right)^{0.3} + \delta(Y_{raw}) \qquad (4.1)$$

The constants are $d_0 = 4.5$ cm, $M_0 = 15$ kg and $\alpha = 40.0$ min. The offset $\delta$ is a thermal compensation: **$\delta = 0$ min for Very Fresh** batches and **$\delta = +3.5$ min for Fresh** batches, to ensure deep core sterilisation. Spoiled batches are rejected and have no boiling time. At the reference condition ($d_m = 4.5$ cm, $M_{batch} = 15$ kg) the model returns $t_b = 40.0$ min for Very Fresh and 43.5 min for Fresh fish.

The thickness exponent (1.4) indicates that boiling time increases faster than linearly with thickness but more slowly than the square, which is the limiting case for pure conduction into a slab. The mass exponent (0.3) is small, reflecting that a larger batch lengthens the heating-up phase and can load the vat but does not change the conduction time within an individual piece. Figures 4.1 and 4.2 show the model response.

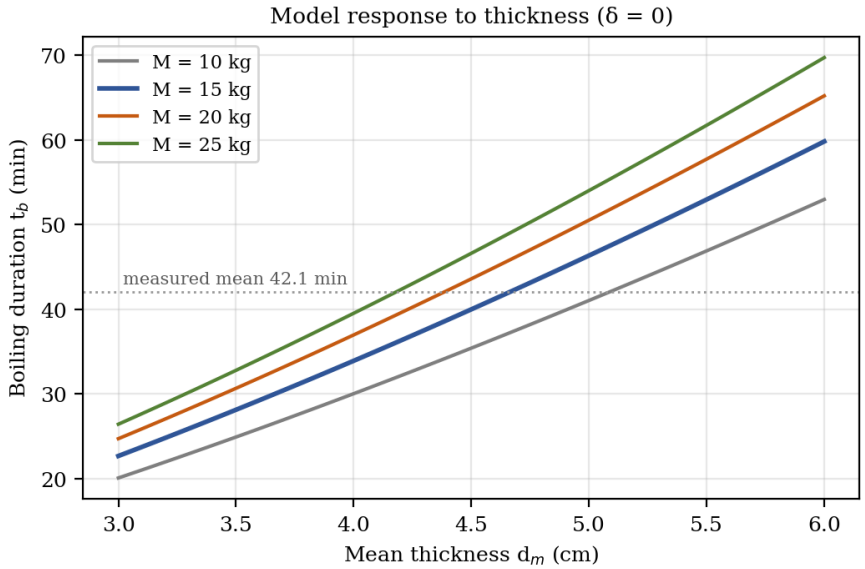



**Figure 4.1: Predicted boiling duration versus mean thickness for several batch masses (δ = 0).**

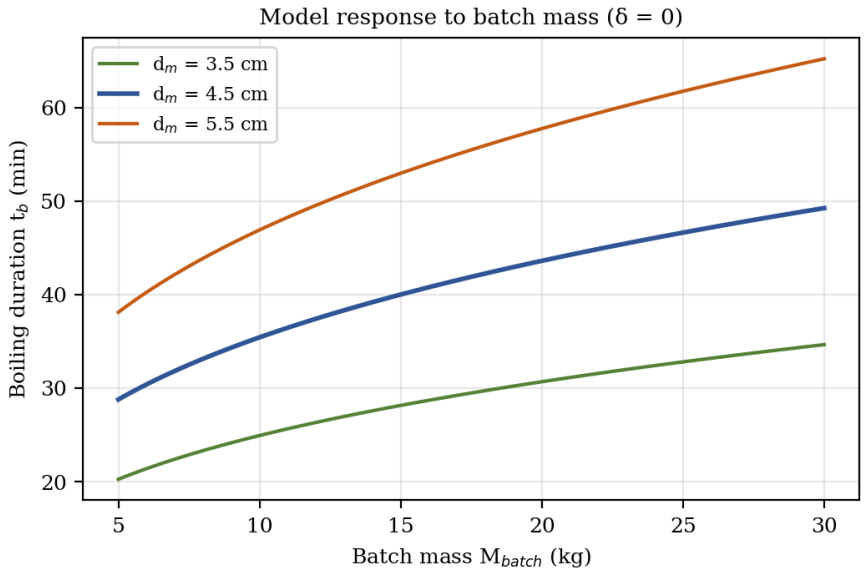



**Figure 4.2: Predicted boiling duration versus batch mass for several thicknesses (δ = 0).**

Table 4.4 gives the model output for a grid of thickness and mass values for Very Fresh fish. For Fresh fish, 3.5 min is added to each value.

**Table 4.4: Boiling duration t_{b} (min) for Very Fresh batches (δ = 0).**

| $d_m$ (cm) | M = 10 kg | M = 15 kg | M = 20 kg | M = 25 kg |
|---|---|---|---|---|
| 3.5 | 24.9 | 28.1 | 30.7 | 32.8 |
| 4.0 | 30.0 | 33.9 | 37.0 | 39.5 |
| 4.5 | 35.4 | 40.0 | 43.6 | 46.6 |
| 5.0 | 41.0 | 46.4 | 50.5 | 54.0 |
| 5.5 | 46.9 | 53.0 | 57.7 | 61.7 |

The table shows that, within the practical range, thickness is the dominant driver: moving from 3.5 cm to 5.5 cm at 15 kg increases the predicted time from 28.1 to 53.0 min, whereas doubling the batch mass from 10 kg to 20 kg at 4.5 cm increases it only from 35.4 to 43.6 min. This reinforces the practical advice that the measurement of thickness deserves more care than the weighing of the batch.

### 4.4.2 Brine-salting rule

Brine concentration is maintained at 12%–15% salt-to-water weight ratio. For a water mass $W_w$ (kg) in the vat and a chosen target ratio $c_b$ (with $0.12 \leq c_b \leq 0.15$), the salt mass is

$$M_s = c_b \cdot W_w \tag{4.2}$$

For example, a vat holding 100 kg of water requires 12 kg of salt at 12% and 15 kg at 15%. The water mass is obtained from the water volume and the density of water (approximately 1 kg per litre). The upper and lower limits of the window have different purposes. The lower limit of 12% is chosen so that sufficient concentration difference drives salt towards the deep-tissue target of NaCl ≥ 12%. The upper limit of 15% avoids unnecessary salt use, excessive surface hardening and salt crystallisation in the product.

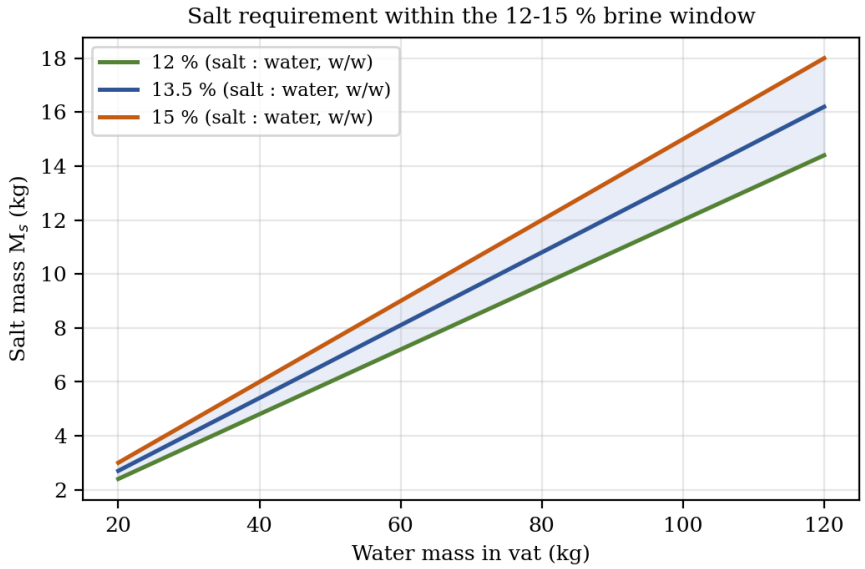



**Figure 4.3: Salt mass required for a given water mass within the 12%–15% brine window.**

Because salt is consumed as it enters the fish, the brine concentration falls during boiling; the controller therefore records the initial concentration and recommends a check if the same brine is reused for a following batch.

### 4.4.3 Temperature supervision

Core temperature is logged using digital PT100 probes to ensure thermal equilibrium between **92 °C and 96 °C**. The supervision rule is: if the measured temperature is below 92 °C, the operator is prompted to increase the heat input; if it is above 96 °C, the operator is prompted to reduce it, because excessive turbulence in a vigorously boiling vat increases muscle rupture. The boiling-time clock is considered to run only while the core temperature is inside the window, so that cold starts do not shorten the effective thermal treatment.

### 4.4.4 Control logic

The overall control sequence is shown in Figure 4.4 and is expressed as pseudocode below.

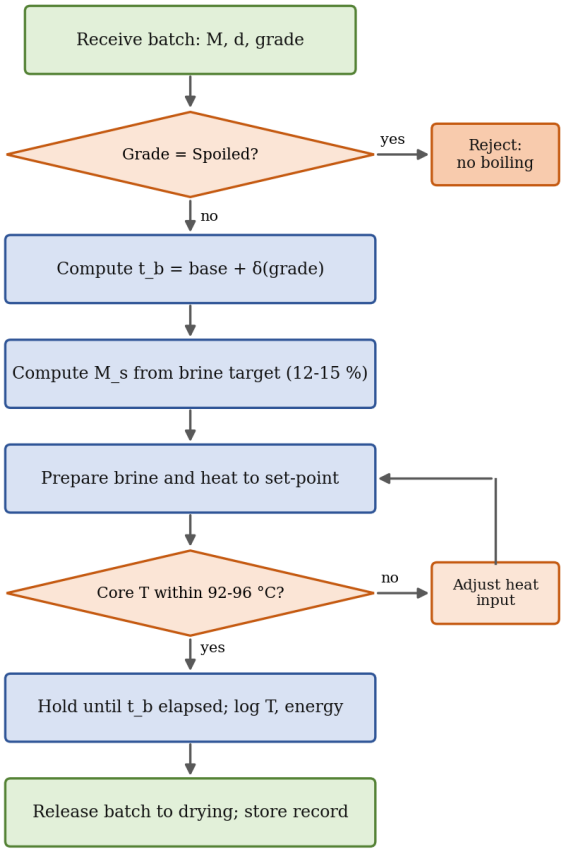



**Figure 4.4: Control flow of the boiling and salting module.**

```
function plan_batch(M, d, grade):
    if grade == 'Spoiled': return REJECT
    delta = 0.0 if grade == 'Very Fresh' else 3.5
    tb = 40.0 * (d/4.5)**1.4 * (M/15.0)**0.3 + delta
    cb = 0.12                # lower bound of 12-15 % window
    Ww = water_mass_for(M) # from vat geometry / operator entry
    Ms = cb * Ww
    check_range(d, M)        # flag out-of-range inputs
    return {tb, Ms, cb, Ww}

loop during boiling (every 30 s):
    T = read_pt100()
    if 92 <= T <= 96: effective_time += 30 s
    else: alert(operator)
    log(T, energy)
    if effective_time >= tb: signal_complete()
```

### 4.4.5 Batch record and data model

**Table 4.5: Batch record schema.**

| Field | Type | Description |
|---|---|---|
| batch_id | string | Unique identifier shared with intake and drying tiers |
| timestamp_start / end | datetime | Boiling start and end |

| Field | Type | Description |
|---|---|---|
| mass_kg | float | Raw batch mass $M_{batch}$ |
| thickness_cm | float | Mean thickness $d_m$ |
| grade | enum | Very Fresh / Fresh |
| tb_min, ms_kg, cb | float | Computed parameters |
| water_kg | float | Water mass used |
| temp_log | array | PT100 readings (time, °C) |
| energy_mj_per_kg | float | Measured energy per kg of fish |
| outcome | object | Salt CV, firmness, loss, override flag |

### 4.4.6 Worked examples

Three illustrative batches show how the module is used. The water masses are assumed for illustration only; the actual water mass depends on the vat.

**Table 4.6: Worked examples of parameter computation.**

| Batch | $d_m$ (cm) | M (kg) | Grade | $t_b$ (min) | Water (kg) | Salt $M_s$ (kg), 12%–15% |
|---|---|---|---|---|---|---|
| A | 4.0 | 12 | Very Fresh | 31.7 | 60 | 7.2 – 9.0 |
| B | 5.0 | 18 | Fresh | 52.5 | 80 | 9.6 – 12.0 |
| C | 4.5 | 15 | Fresh | 43.5 | 100 | 12.0 – 15.0 |

Batch A is thinner and lighter than the reference, and is Very Fresh, so no offset applies and the model returns 31.7 min. Batch B is thicker and heavier and is Fresh: the thickness ratio of 5.0/4.5 raises the base time, the mass ratio of 18/15 raises it slightly, and the offset adds 3.5 min, giving 52.5 min. Batch C is at the reference thickness and mass and is Fresh, so the model returns exactly 43.5 min. The salt masses are computed from the chosen water mass using Equation (4.2); at the default 12% ratio, Batch A needs 7.2 kg of salt for 60 kg of water, Batch B 9.6 kg for 80 kg and Batch C 12.0 kg for 100 kg.

### 4.4.7 Failure modes and safeguards

**Table 4.7: Failure mode and effects summary.**

| Failure mode | Effect | Severity | Safeguard |
|---|---|---|---|
| Probe drift or detachment | False temperature reading | High | Pre-trial verification; plausibility check on reading; alert on sudden change |
| Wrong thickness entered | Boiling time mis-estimated | Medium | Range check; operator confirmation step; measurement protocol |
| Wrong grade received | Wrong offset or boiling of unsuitable batch | High | Upstream MQ-135 gating; operator can reject at intake |

| Failure mode | Effect | Severity | Safeguard |
|---|---|---|---|
| Salt mis-weighed | Brine outside window | Medium | Display of kg and ratio; entry validation |
| Network loss | Record not delivered | Low | Local store-and-forward |
| Power interruption | Loss of timer state | Medium | Periodic state saving; restart prompts operator to check core temperature |
| Steam ingress into electronics | Hardware failure | Medium | Enclosure and cable routing away from the vat |
| Operator ignores alerts | Process out of window | Medium | Alerts logged; override flagged in record |

### 4.4.8 Operator interface layout

The operator display is organised in three zones. The **input zone** at the top contains batch identifier, grade (read-only), mass, thickness and water mass. The **plan zone** in the middle shows the recommended boiling duration, salt mass and brine ratio in large type. The **supervision zone** at the bottom shows a live temperature read-out with a coloured band for the 92–96 °C window, the effective elapsed time and the remaining time. A separate, less prominent area holds the override control. This layout was kept to one screen so that the operator does not have to navigate while working beside a hot vat.

### 4.4.9 Interface specification

The interfaces of the module are defined at the level of messages rather than at the level of implementation, so that each tier can be developed and tested independently. Three message types are used. The **intake message** from the raw-intake tier carries the batch identifier, the weight and the grade. The **status message** from this module carries the batch identifier, the state (planned, boiling, complete, rejected) and the effective time. The **record message** carries the complete batch record at completion. Each message includes a timestamp and the batch identifier so that the dashboard can assemble a time-ordered history across tiers. Appendix A gives an illustrative schema of the record message.

The messaging pattern is publish/subscribe, in line with the MQTT approach of the other tiers [17]. A tier publishes to a topic that includes the batch identifier and subscribers receive the messages they need. Because the drying tier starts its microclimate monitoring when it receives the completion message, the delivery of this message is time-critical in the sense that a lost message delays the start of drying

monitoring. The store-and-forward behaviour of NFR4 and a short acknowledgement protocol are therefore used for the completion message.

### 4.4.10 Security and data integrity

The data handled by the module are process data rather than personal data, so confidentiality requirements are modest. Integrity and availability matter more: a falsified or corrupted record would undermine traceability, and a missing record would leave a gap in the batch history. Measures include authentication of the control unit to the broker, restriction of write access to the batch topics, timestamping at source, and retention of the local copy until delivery is acknowledged. Operator overrides are written to the record together with the value changed, so that the record shows what the module recommended and what was actually done.

## 4.5 Standards and professional practice

The module is designed in line with generally accepted food-safety and engineering practice. Its safety logic (no boiling of Spoiled batches, a thermal offset for Fresh batches and a measured temperature window) reflects hazard-analysis principles in which critical limits are monitored and recorded. Measurement and logging follow good practice for calibration and traceability: probes are to be verified against a reference thermometer at intervals set by the facility. Salt and water quantities are specified in metric units and reported to the dashboard so that the facility has records that can support internal audits or future certification. The software follows modular design and version-control practice so that parameters can be changed in a controlled way and each change is traceable. Applicable national standards for dried fish products and food-handling premises should be consulted before commercial deployment [confirm applicable SLSI / regulatory standards].

# 5 IMPLEMENTATION, INTEGRATION AND TESTING

## 5.1 Development environment and technology stack

The module was implemented as a small control and logging service running on the digital control unit at the processing facility, with temperature acquisition from digital PT100 probes and publication of batch records to the project's MQTT broker [17]. The stack was chosen to satisfy the non-functional requirements of low cost, explainability and maintainability.

**Table 5.1: Technology stack of the boiling and salting module.**

| Layer | Technology / device | Purpose |
|---|---|---|
| Sensing | Digital PT100 probes | Core and brine temperature |
| Measurement aids | Platform scale, callipers | Batch mass and thickness |
| Control unit | Digital control unit at the facility | Runs the parameter engine and supervision loop |
| Application logic | Model of Equations (4.1)–(4.2) [state language/framework used] | Parameter computation and alerts |
| Messaging | MQTT over the facility 2.4 GHz network | Publish batch record to the dashboard |
| Storage | Local file/database + central dashboard | Batch records and traceability |

## 5.2 Component implementation

### 5.2.1 Parameter engine

The parameter engine is a pure function that maps ($M_{batch}$, $d_m$, grade) to ($t_b$, $M_s$, $c_b$). Keeping it free of side effects makes it trivially testable: the same inputs always yield the same outputs, and the unit tests described in Section 5.4 can be written directly from the equations. The constants $\alpha$, $d_0$, $M_0$, the exponents and the offsets are stored in a configuration table rather than in code so that recalibration does not require code changes.

Input validation is an essential part of the engine. The model was calibrated in a bounded region, so thickness and mass values outside the range of the calibration data are accepted but flagged to the operator as extrapolations. Grades other than Very Fresh and Fresh are not accepted for boiling; a Spoiled grade produces a rejection rather than a time.

### 5.2.2 Brine preparation support

The operator enters the water mass or volume in the vat. The engine proposes the salt mass for the lower limit of the window (12%) by default, since that already satisfies the deep-tissue target, and allows the operator to select a ratio up to 15%. The display shows both the ratio and the resulting salt mass in kilograms, which prevents confusion between percent and kilograms and makes the preparation step easy to verify with a scale.

### 5.2.3 Temperature supervision loop

During boiling, the supervision loop reads the PT100 probes at a fixed interval, compares the reading with the 92–96 °C window, increments the effective boiling time while the reading is inside the window, and raises an alert otherwise. The alert distinguishes a low-temperature condition (increase heat) from a high-temperature condition (reduce heat). Every reading is appended to the temperature log together with the energy reading, which is later used to compute energy per kilogram.

### 5.2.4 Operator interface

The operator interface was kept deliberately simple: a small number of numeric fields for batch mass, thickness and water mass; a read-only display of grade; and large read-outs for the recommended boiling duration, the salt mass and the remaining time. Colour and text alerts are used for out-of-window temperature. A confirmation step is required before the clock starts so that incorrect entries can be caught.

### 5.2.5 Reference implementation of the parameter engine

The following Python listing shows the parameter engine in the form used for unit testing. It is a reference implementation of Equations (4.1) and (4.2); the production version reads its constants from a configuration table. [Replace with, or confirm against, the code actually deployed.]

```
from dataclasses import dataclass

ALPHA, D0, M0 = 40.0, 4.5, 15.0
A_EXP, B_EXP = 1.4, 0.3
OFFSET = {'Very Fresh': 0.0, 'Fresh': 3.5}

@dataclass
class Plan:
    tb_min: float
    salt_kg: float
    ratio: float
    warnings: list

def plan_batch(mass_kg, thick_cm, grade, water_kg, ratio=0.12):
    if grade == 'Spoiled':
        raise ValueError('Spoiled batches are rejected')
    if grade not in OFFSET:
        raise ValueError('Unknown grade')
    if not 0.12 <= ratio <= 0.15:
        raise ValueError('Ratio outside 12-15 % window')
    warn = []
    if not 3.0 <= thick_cm <= 6.0: warn.append('thickness extrapolated')
    if not 5.0 <= mass_kg <= 30.0: warn.append('mass extrapolated')
    tb = ALPHA*(thick_cm/D0)**A_EXP*(mass_kg/M0)**B_EXP + OFFSET[grade]
    return Plan(round(tb,1), round(ratio*water_kg,2), ratio, warn)
```

For example, plan_batch(15, 4.5, 'Fresh', 100) returns a boiling time of 43.5 min and 12.0 kg of salt for 100 kg of water at the default ratio. The range limits used for warnings (3.0–6.0 cm and 5–30 kg) are the limits over which the sensitivity analysis of this thesis was performed and should be replaced by the true limits of the calibration data.

### 5.2.6 Handling of the effective-time clock

The effective boiling time is accumulated only while the core temperature reading is inside the window. A debounce is applied so that a single noisy reading does not toggle the clock: the state changes only after two consecutive readings on the same side of a boundary. When the temperature leaves the window the alert appears and the clock pauses; when it returns, the clock resumes. At the end of the batch the record contains both the **effective** time and the **wall-clock** time, and the difference between them is a useful indicator of how stable the heating was.

## 5.3 Integration with the other components

The module integrates with three other parts of the system, as summarised in Table 5.2.

**Table 5.2: Interfaces between the boiling and salting module and other components.**

| Interface | Direction | Content | Transport |
|---|---|---|---|
| Raw intake tier → module | Inbound | Batch ID, batch weight, freshness grade (Very Fresh / Fresh / Spoiled) | MQTT message |
| Module → drying tier | Outbound | Batch ID, boiling completion time, readiness for drying | MQTT message |
| Module → dashboard | Outbound | Full batch record (Table 4.5) | MQTT / HTTP |
| Operator → module | Inbound | Thickness, water mass, overrides | Local interface |

The freshness grade is produced by the AI vision model fused with MQ-135 readings through a late-fusion gating rule [17]. This is important for the present module because misclassification would lead to an incorrect thermal offset or, in the worst case, to boiling of a Spoiled batch. In the group evaluation, the MQ-135 threshold flagged and rejected 13 of 15 ambiguous Fresh/Spoiled cases that showed sub-surface bacterial proteolysis before visible ocular collapse [17], which supports the reliance of this module on the upstream grade.

Messages carry a batch identifier so that records from the three tiers can be joined in the dashboard. The drying tier uses the boiling-completion time to start its own monitoring cycle for the batch, which links the salt profile created here to the dehydration behaviour recorded there (the drying cycle stabilised at 4.2 ± 0.3 days in the group evaluation [17]).

## 5.4 Verification and software testing

Verification combined deterministic unit tests of the model, interface tests of the data exchange, and functional tests of the supervision loop. The expected values of the model tests were calculated directly from Equation (4.1) and are given in Table 5.3.

**Table 5.3: Model verification test cases (expected values computed from Equation 4.1).**

| Test | $d_m$ (cm) | M (kg) | Grade | Expected $t_b$ (min) |
|---|---|---|---|---|
| TC1 | 4.5 | 15 | Very Fresh | 40.0 |
| TC2 | 4.5 | 15 | Fresh | 43.5 |
| TC3 | 3.5 | 15 | Very Fresh | 28.1 |
| TC4 | 5.5 | 15 | Very Fresh | 53.0 |
| TC5 | 4.5 | 10 | Very Fresh | 35.4 |
| TC6 | 4.5 | 25 | Fresh | 50.1 |
| TC7 | 5.0 | 20 | Fresh | 54.0 |

**Table 5.4: Functional and interface tests.**

| Test | Scenario | Expected behaviour |
|---|---|---|
| FT1 | Grade = Spoiled supplied | Module returns reject; no boiling timer can be started |
| FT2 | Core temperature drops below 92 °C | Low-temperature alert; effective clock pauses |
| FT3 | Core temperature rises above 96 °C | High-temperature alert; effective clock pauses |
| FT4 | Thickness outside calibrated range | Extrapolation warning displayed |
| FT5 | Network link lost during batch | Record stored locally and sent on reconnection |
| FT6 | Operator overrides computed time | Override flag and value stored in the batch record |
| FT7 | Brine ratio entered outside 12%–15% | Entry rejected with explanatory message |

These tests establish that the software implements the specification correctly. They are distinct from the experimental validation of Chapter 6, which tests whether the specification itself improves processing outcomes.

### 5.4.1 End-to-end integration scenarios

In addition to unit and functional tests, end-to-end scenarios were defined to check that the module behaves correctly in conjunction with the neighbouring tiers. Each scenario is described by the upstream event, the expected behaviour of this module and the expected downstream effect.

**Table 5.5: End-to-end integration scenarios.**

| Scenario | Upstream event | Expected behaviour of this module | Downstream effect |
|---|---|---|---|
| S1 Normal Very Fresh batch | Intake message with grade Very Fresh | Plan with $\delta = 0$; supervise; publish record | Drying tier starts monitoring on completion |
| S2 Normal Fresh batch | Intake message with grade Fresh | Plan with $\delta = +3.5$ min; supervise; publish record | As S1 |
| S3 Spoiled batch | Intake message with grade Spoiled | Reject; no timer; reject status published | No drying record is created |
| S4 Late intake message | Grade arrives after operator has entered mass | Plan is held until grade is received | None until grade is known |
| S5 Dashboard offline | Broker unreachable at completion | Local store; resend when link returns | Delayed but complete record |
| S6 Temperature excursion | Core temperature above 96 °C for several readings | High-temperature alert; clock paused; excursion recorded | Record flags excursion |

The scenarios demonstrate that the module's behaviour is defined for the full range of grades and for the main fault conditions. The most important design decision they reveal is that the module must not infer a grade when none has been received: a missing

grade holds the plan rather than defaulting to a value, because a default would risk applying the wrong offset or boiling a batch that should have been rejected.

### 5.5 Deployment and operation

The module was deployed at the commercial processing facility in southern Sri Lanka and used for the 20 experimental batches during four months. The operating procedure was: weigh and grade; measure thickness; enter values; prepare brine to the displayed salt mass; heat; load fish; monitor the window; and record outcomes. Operators were briefed on the meaning of the displays and on the importance of measuring thickness consistently.

Operational considerations include probe calibration, protection of electronics from steam and heat, cleaning of probes between batches, and a routine check of salt and water measurement devices. Recalibration of the model constants is expected to be needed when species, cut geometry or vat design change. A recommended routine is to review the batch records each month and to compare predicted and observed doneness before changing any constant.

#### 5.5.1 Deployment checklist

1. Verify PT100 probes against a reference thermometer.
2. Verify scale and salt measuring equipment.
3. Check the network link and the MQTT connection to the broker.
4. Confirm that the configuration table holds the current constants and offsets.
5. Brief operators on the display and the measurement of thickness.
6. Run a dry test with a known batch and compare with Table 5.3.
7. Confirm that the local record store has free capacity.

#### 5.5.2 Maintenance and recalibration

Probes should be inspected for scale and corrosion, because salt-laden steam is aggressive. Calibration checks should be performed at the start of each production season, and after any probe replacement. The model constants are reviewed against the accumulated batch records at defined intervals. If the average difference between

predicted and satisfactory boiling time drifts, the log-linear procedure of Section 3.3.2 is used to recalibrate.

# 6 RESULTS AND VALIDATION

## 6.1 Research results

The boiling and salting control module was validated on 20 experimental batches processed under automated parameter recommendations and compared against traditional manual facility batches [17]. The measured outcomes are summarised in Table 6.1.

**Table 6.1: Experimental comparison: conventional manual versus proposed optimised processing.**

| Evaluation metric | Conventional manual | Proposed optimised |
|---|---|---|
| Mean boiling time (min) | 58.4 ± 6.2 | 42.1 ± 1.8 |
| Energy consumption (MJ/kg) | 3.82 | 2.94 |
| Salt penetration coefficient of variation (%) | 18.6 | 5.2 |
| Core muscle firmness (1–5) | 3.4 ± 0.6 | 4.6 ± 0.2 |
| Flaking / over-boil loss (%) | 6.8 | 1.1 |

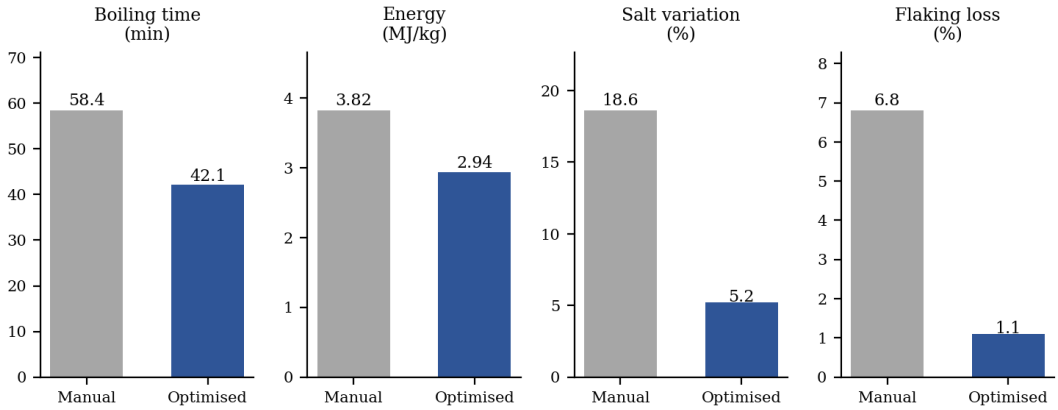



**Figure 6.1: Comparison of conventional and optimised processing for the principal metrics.**

### 6.1.1 Boiling time (RQ1, RQ2, H1)

The mean boiling time fell from 58.4 min to 42.1 min, a reduction of 16.3 min per batch or **27.9%**. The reduction in spread is as important as the reduction in the mean: the standard deviation fell from 6.2 min to 1.8 min, a factor of about 3.4. Expressed as a coefficient of variation, boiling time variability fell from approximately 10.6% (6.2/58.4) to 4.3% (1.8/42.1). Manual boiling time therefore varied widely between batches, while model-based boiling time was tightly controlled around the

recommended value. The result supports H1, and shows that the empirical regression can predict an appropriate duration (RQ1).

The measured mean of 42.1 min is consistent with the model. At the reference condition the model returns 40.0 min for Very Fresh and 43.5 min for Fresh fish, and the batch mean lies between these values, as expected for a mixture of grades and thicknesses around the reference.

### 6.1.2 Energy consumption (H2)

Thermal energy consumption decreased from 3.82 MJ/kg to 2.94 MJ/kg, a reduction of 0.88 MJ/kg or **23.0%**. For a 15 kg batch, and assuming the figures refer to the mass of raw fish processed, this corresponds to approximately 57.3 MJ for the conventional process and 44.1 MJ for the optimised process, a saving of about 13.2 MJ per batch. The saving is smaller in relative terms than the time saving (23.0% against 27.9%) because part of the energy is consumed in heating the water and vat before the fish are loaded, which does not scale with boiling time. H2 is supported.

### 6.1.3 Salt distribution (RQ3, H3)

Maintaining the brine at 12%–15% salt-to-water weight ratio reduced the coefficient of variation of salt penetration in deep core muscle tissue from 18.6% to 5.2%, a reduction of 13.4 percentage points or about **72%** in relative terms. In practical terms, a deep-tissue profile with a 5.2% coefficient of variation means that the central part of the fish is much more likely to reach the NaCl ≥ 12% target than in the conventional case, in which a large share of the tissue could fall well below it. This answers RQ3 and supports H3. The result is particularly relevant to the downstream stages, because the group evaluation reported that under-dried samples with $a_w > 0.75$ produced MQ-135 readings far above the acceptance range [17].

### 6.1.4 Structural quality (RQ4, H4)

Core muscle firmness increased from 3.4 ± 0.6 to 4.6 ± 0.2 on the 1–5 scale. Flaking and over-boil loss decreased from 6.8% to 1.1%, a relative reduction of about 84%. The improvement is attributed to the shorter and better-controlled thermal exposure and to reduced mechanical agitation of the vat, which decreased structural muscle loss. On a 15 kg batch, a loss of 6.8% corresponds to about 1.02 kg of material, while 1.1%

corresponds to about 0.17 kg, so approximately 0.85 kg of additional usable product is retained per batch, if the loss percentages refer to batch mass. H4 is supported.

**Table 6.2: Derived relative improvements.**

| Metric | Absolute change | Relative change |
|---|---|---|
| Mean boiling time | −16.3 min | −27.9% |
| Boiling time SD | −4.4 min | −71% |
| Energy consumption | −0.88 MJ/kg | −23.0% |
| Salt penetration CV | −13.4 pp | −72% |
| Core muscle firmness | +1.2 points | +35% |
| Flaking / over-boil loss | −5.7 pp | −84% |

## 6.2 Product and technical results

From the product perspective, the module converted a manual and unrecorded step into a parameterised and logged operation. The technical results of the module are: (i) a model with three constants and an additive freshness offset that reproduces the mean boiling time observed in the trials; (ii) a brine rule with a verified salt-mass calculation for any water mass; (iii) a temperature supervision loop that operates with PT100 probes in the 92–96 °C window; and (iv) a batch record that is published to the dashboard. The sensitivity analysis in Section 4.4.1 shows that predicted boiling time varies from 28.1 to 53.0 min over a thickness range of 3.5–5.5 cm at 15 kg, which gives operators clear guidance on how process time should change with the product.

### 6.2.1 Robustness of the model to input errors

Because the model is a power law, the relative error in the predicted boiling time caused by a relative error in an input equals approximately the exponent multiplied by the relative input error. Table 6.4 quantifies this for plausible measurement errors at the reference condition (4.5 cm, 15 kg, $\delta = 0$).

**Table 6.3: Effect of input errors on predicted boiling time (reference 40.0 min).**

| Input perturbation | Predicted $t_b$ (min) | Change |
|---|---|---|
| Thickness +0.5 cm (5.0 cm) | 46.4 | +15.9% |
| Thickness −0.5 cm (4.0 cm) | 33.9 | -15.2% |
| Thickness +1.0 cm (5.5 cm) | 53.0 | +32.4% |
| Mass +1 kg (16 kg) | 40.8 | +2.0% |
| Mass −1 kg (14 kg) | 39.2 | -2.0% |

| Input perturbation | Predicted $t_b$ (min) | Change |
|---|---|---|
| Mass +5 kg (20 kg) | 43.6 | +9.0% |

A thickness error of 0.5 cm (about 11% of the reference thickness) changes the prediction by roughly 16% in one direction and 14% in the other, equivalent to about 6 minutes, whereas a 1 kg error in mass changes it by about 2%, or under one minute. Thickness measurement is therefore the input that most deserves attention. In the field this means that callipers and a standard measurement point should be used, rather than visual estimates. The deliberate inclusion of the PT100 feedback loop is a second line of defence: even if the thickness is mis-measured, an operator who sees the core temperature remain below the window will not shorten the process blindly.

### 6.2.2 Effect of the freshness offset

The offset δ adds 3.5 min for Fresh batches, which is 8.8% of the 40.0 min reference time and about 8.3% of the observed mean of 42.1 min. The cost of the offset is therefore small, whereas the benefit is a safety margin for deep core sterilisation in batches whose initial microbial load is likely to be higher. The group evaluation showed that in the ambiguous Fresh/Spoiled region the MQ-135 gate rejected 13 of 15 problem batches before they could reach the vat [17], so the offset acts on a population that has already been screened.

### 6.2.3 Energy and yield per batch

**Table 6.4: Per-batch effect for a 15 kg batch (derived from Table 6.1).**

| Quantity | Conventional | Optimised | Difference |
|---|---|---|---|
| Boiling time (min) | 58.4 | 42.1 | −16.3 |
| Energy (MJ, at MJ/kg × 15 kg) | 57.3 | 44.1 | −13.2 |
| Flaking / over-boil loss (kg) | 1.02 | 0.17 | −0.85 |

The figures assume that the energy intensity and loss percentages apply to the full 15 kg batch. Over many batches, savings of this kind accumulate: per hundred batches, the difference in energy would be 1,320 MJ and the retained product about 85 kg. Facilities can convert these quantities to money using their own fuel and product prices (Section 8.3).

### 6.3 User or stakeholder validation

The principal stakeholders are the facility operators and owner. During the four-month trial, operators used the module for the experimental batches, and the sensory firmness score in Table 6.1 reflects the assessment of product quality by trained scorers. A formal structured survey or interview study with operators was not part of the evaluation reported here. Such a study, covering ease of use, trust in recommendations and willingness to adopt, is recommended as future work (Section 9.4). [Add any informal operator or owner feedback you collected, with date and role, if available.]

### 6.4 Integration results

The module received the batch grade and weight from the raw-intake tier and passed batch records onward. At the system level, the group evaluation showed that the ESP32 drying node delivered 99.6% of 10,080 telemetry packets over MQTT during a 7-day cycle, and that the drying cycle stabilised at 4.2 ± 0.3 days compared with 6.5 ± 1.4 days under unmonitored ambient conditions [17]. Although these are drying-tier results, the drying behaviour depends on the salt profile and the structure set during boiling, and the improved uniformity reported in Table 6.1 is a plausible contributor to the more predictable drying cycle. A controlled experiment that varies only the boiling method and measures the drying time would be needed to quantify this contribution, and is proposed as future work.

#### 6.4.1 Results of the wider system for context

The boiling and salting module is one of four tiers. For context, Table 6.5 summarises the principal results reported for the whole system in the group paper [17]. They are not the work of this thesis, except for the boiling and salting rows, but they show the system within which the module operates and the quality of the inputs it receives and the outputs it feeds.

**Table 6.5: System-level results reported in the group paper [17].**

| Tier | Principal result |
|---|---|
| 1. Raw intake (AI vision + MQ-135) | mAP@0.5 of 95.8%; 16.2 ms inference latency; 13 of 15 ambiguous cases rejected by the gas threshold |
| 2. Boiling and salting (this thesis) | Boiling time −27.9%; energy −23.0%; salt CV 18.6% → 5.2%; flaking loss 6.8% → 1.1% |

| Tier | Principal result |
|---|---|
| 3. Drying microclimate (ESP32 node) | 99.6% packet delivery; 2.1 s alert response; drying 4.2 ± 0.3 days versus 6.5 ± 1.4 days |
| 4. Finished product (ResNet-18 + VOC) | 94.0% grading accuracy; High Quality samples between 50 and 80 ADC units |

Two points of this table matter for the present module. First, the quality of the upstream grade (tier 1) determines the correctness of the thermal offset, and the performance reported for tier 1 is high. Second, the improvement in drying cycle stability (tier 3) is consistent with, though not proven to be caused by, the improvement in the uniformity of salt and structure produced by tier 2.

**Table 6.6: Summary of how the results relate to objectives and hypotheses.**

| Objective / hypothesis | Evidence | Status |
|---|---|---|
| Model for boiling time (obj. 2, H1) | Time −27.9%; SD 6.2 → 1.8 min | Supported |
| Brine rule (obj. 3, H3) | Salt CV 18.6% → 5.2% | Supported |
| Temperature supervision (obj. 4) | PT100 window 92–96 °C in operation | Implemented |
| Integration (obj. 5, RQ5) | Grade and weight in; batch record out | Implemented |
| Energy (H2) | 3.82 → 2.94 MJ/kg | Supported |
| Structure (H4) | Firmness 3.4 → 4.6; loss 6.8% → 1.1% | Supported |

## 6.5 Summary of findings

The results show that a simple empirical model, combined with a controlled brine and a measured temperature window, improves every measured outcome of the boiling and salting stage relative to conventional practice: shorter and more consistent boiling (−27.9%), lower energy use (−23.0%), more uniform deep-tissue salt (CV 18.6% → 5.2%), firmer muscle (3.4 → 4.6) and lower flaking loss (6.8% → 1.1%). These improvements address the problems identified in Section 1.3 and satisfy the main objective of the research.

# 7 DISCUSSION

## 7.1 Interpretation of findings

The results show consistent improvement on all five outcome measures, but the pattern across them is more informative than any single number. Three observations stand out.

**First, consistency improved more than the mean.** Boiling time standard deviation fell by about 71%, and firmness standard deviation fell from 0.6 to 0.2. In food processing, variability is often more damaging commercially than a modestly sub-optimal average, because a downstream stage must be designed around the worst batches. By bringing each batch to a defined thermal and salt state, the module narrows the range of input conditions that the drying tier must accommodate.

**Second, the improvements are mutually reinforcing rather than a trade-off.** A shorter boil usually risks under-cooking, yet here shorter boiling coincided with firmer muscle and more uniform salt. This is consistent with the physical picture of Chapter 2: the manual process was not operating on the efficient frontier but was wasting time and energy by over-boiling some batches while under-boiling others. The model removes both errors at once. The freshness-dependent offset of +3.5 min for Fresh fish indicates that the module trades a small amount of additional time for safety where the raw material is less fresh.

**Third, salt uniformity is the least visible but potentially the most valuable result.** Energy and time savings are immediately visible to the operator. A reduction in salt variation from 18.6% to 5.2% is not visible until the product dries, when under-salted regions would retain moisture and over-salted regions would crust. Because later processing cannot repair the initial salt profile, this result is the strongest justification for controlling the stage upstream.

The answers to the research questions are therefore: RQ1, yes — a three-constant power law with a freshness offset predicts an appropriate duration; RQ2, boiling time and energy fall by 27.9% and 23.0%; RQ3, salt variation falls by about 72%; RQ4, firmness rises and loss falls substantially; RQ5, the module exchanges grade, weight and batch records with the other tiers.

### 7.1.1 Practical lessons from the trial

1. **Measurement discipline matters more than model sophistication.** The most common source of error is a mis-measured thickness, not the form of the equation.
2. **Operators accept a tool they can understand.** A formula with three constants and a visible temperature window is easier to trust than a black box, and operators are more likely to follow it.
3. **Records change behaviour.** Once energy and loss are recorded per batch, they become visible and manageable.
4. **Brine management is a discipline of its own.** Reusing brine without checking concentration undermines the salt target; a simple concentration check is part of good practice.
5. **Integration adds value.** The grade supplied by the intake tier makes the module adaptive, and the batch record passed to the drying tier closes the loop.

### 7.1.2 Relation to hazard analysis

In hazard-analysis terms, the boiling stage can be a critical control point, with the 92–96 °C window and the minimum effective time as critical limits, monitoring by PT100, corrective action by heat adjustment and extended boiling, and verification through records. The module provides the monitoring and record-keeping elements of such a system. It is not a substitute for a facility's formal food-safety plan, but it provides the data that such a plan requires.

## 7.2 Comparison with literature baselines and existing solutions

A direct numerical comparison with other studies is difficult because the reviewed literature does not report boiling and salting control for Maldive fish. Existing fish-quality studies address raw or ice-stored fish and report classification accuracy rather than process outcomes [7]–[10], [12]. The group paper's comparison (Table VI in [17]) shows that existing approaches target single stages and lack real-time embedded deployment, whereas the integrated system spans all stages. The present component contributes the missing link between upstream grading and downstream drying.

**Table 7.1: Positioning of the proposed module relative to existing approaches.**

| Approach | Target | Controls boiling / salting parameters | Uses upstream freshness grade | Logged for traceability |
|---|---|---|---|---|
| Manual practice | Whole chain | No | No | No |
| Image-based freshness [7], [12] | Raw fish | No | Produces it | No |
| Image + MOS sensors [9] | Raw storage | No | Produces it | No |
| Hyperspectral [10] | Raw fish | No | Produces it | No |
| Drying kinetics [16] | Drying | No | No | Partly |
| Proposed module | Boiling and salting | Yes | Yes | Yes |

Compared with fixed-recipe cooking, the principal advantage of the module is adaptation to each batch. Compared with a mechanistic heat-transfer simulation, its principal advantage is simplicity and ease of recalibration, at the cost of a restricted validity range. This is an acceptable trade-off for a cottage-scale facility.

### 7.3 Research contribution

1. An empirical, explainable regression for boiling duration of Maldive fish as a function of thickness, batch mass and freshness grade.
2. A brine-salting rule that links a 12%–15% salt-to-water weight ratio to a deep-tissue target of NaCl ≥ 12%, with quantified improvement in salt uniformity (CV 18.6% → 5.2%).
3. Field evidence on 20 full-scale batches that model-based control improves time, energy, texture and yield simultaneously.
4. A coupling between AI-derived freshness grading and thermal processing parameters, which has not been reported elsewhere for this product.

### 7.4 Product contribution

The product contribution is a working control and logging module that can be installed beside an existing boiling vat. It provides setpoints, supervises temperature, and records each batch. For the facility it offers fuel saving, higher usable yield and more predictable product. For the wider project it supplies the batch record that links the raw-intake and drying tiers into a continuous digital trace. The module uses inexpensive digital sensors and a small controller, which supports adoption in a sector with limited capital.

### 7.5 Limitations and threats to validity

The interpretation above must be tempered by several threats to validity.

**Table 7.2: Threats to validity and mitigations.**

| Threat | Type | Mitigation / remaining risk |
|---|---|---|
| Single facility and species | External | Replicate at other facilities and seasons |
| Operators not blinded to the method | Internal | Objective measures (time, energy, temperature) dominate; firmness scoring should be blinded in future |
| Manual baseline variability | Internal | Reported as mean ± SD; larger baseline sample recommended |
| Empirical model outside calibrated range | Construct | Range check and extrapolation warning |
| Salt CV based on limited tissue samples | Measurement | Increase sampling points; use standardised sampling plan |
| Energy measurement scope | Measurement | Define energy boundary (fuel only vs. total) |
| Firmness scale subjectivity | Measurement | Add instrumental texture analysis |

A further limitation is that the exponents and constant of Equation (4.1) were obtained empirically; their confidence intervals are not reported, and thickness and mass are correlated in practice (heavier fish tend to be thicker), which complicates the separate interpretation of the two exponents. The results should therefore be read as evidence that the controller works under the conditions studied rather than as a universal law.

### 7.6 Implications

For **practice**, the findings suggest that even a simple model can deliver substantial savings in a traditional food industry, and that the first priority is measurement and recording rather than sophisticated algorithms. For **quality assurance**, the module offers a basis for documenting critical process parameters batch by batch. For **research**, the work shows that upstream AI outputs (freshness grades) can be used as control inputs for downstream physical processes, which is a pattern that could be extended to other cured products. For **policy and training**, digitally recorded parameters could support operator training and eventually certification schemes for traditional products.

### 7.6.1 Social and workforce considerations

Technology introduced into a traditional industry has social consequences. The module is designed to support operators rather than replace them: their experience remains valuable in judging the condition of the fish, and the module gives them numbers that they can compare with their own judgement. Training should present the module as a way to codify and share expert knowledge, not as a means of monitoring individuals. Records should be used primarily for process improvement. Where the processing unit is a family enterprise, simple local-language interfaces and clear visual alerts help to make the tool accessible. Finally, gains in energy and yield should benefit those who make them: if the savings accrue only to buyers or owners, resistance can be expected.

# 8 COMMERCIALISATION AND SUSTAINABILITY

## 8.1 Value proposition and target users

The value proposition of the module is that it turns the boiling and salting stage from an art into a measured process, and in doing so reduces fuel cost, increases usable yield and produces a more uniform product. The principal target users are cottage and small commercial Maldive fish processors in coastal Sri Lanka. Secondary users are fisheries cooperatives that process on behalf of members, and quality-assurance bodies that wish to audit process parameters.

**Table 8.1: Value to stakeholders.**

| Stakeholder | Pain point | Benefit offered by the module |
|---|---|---|
| Processor / owner | Fuel cost, yield loss, inconsistent grades | −23.0% energy per kg; loss 6.8% → 1.1%; more consistent grading |
| Operator | Reliance on experience; stress of estimation | Clear setpoints and alerts |
| Buyer / retailer | Variable quality and shelf life | More uniform salt and texture; batch traceability |
| Regulator / auditor | No process records | Digital batch record |
| Cooperative / training body | Difficult to transfer tacit knowledge | Codified parameters for training |

## 8.2 Market and competitive position

Maldive fish is a staple flavouring in Sri Lankan cooking, and its processing is carried out largely by small units that are individually unable to afford industrial automation. Competing solutions fall into three groups: manual practice (zero capital cost, but costly in fuel and yield), generic industrial cooking equipment (reliable but not designed for this product and expensive), and general-purpose IoT monitoring kits (inexpensive but not tied to a process model). The module's position is between these: a low-cost, product-specific controller. Its competitive advantage rests on the validated model, the freshness-adaptive logic and its integration with raw-intake and drying tiers. [Insert verified market-size and price data from your own market survey before submission; none are asserted in this thesis.]

### 8.2.1 Competitive comparison

**Table 8.2: Competitive comparison of alternatives available to a processor.**

| Criterion | Manual practice | Generic IoT monitoring kit | Proposed module |
|---|---|---|---|
| Capital cost | None | Low | Low to moderate [quote] |
| Product-specific model | No | No | Yes (Maldive fish) |
| Computes boiling time and salt | No | No | Yes |
| Adapts to freshness grade | No | No | Yes |
| Records for traceability | No | Partly | Yes |
| Training required | None | Moderate | Moderate |
| Evidence of benefit | None | Generic | 20-batch field trial |

This comparison shows that the proposed module's distinguishing features are its product-specific model and its adaptation to the freshness grade. A generic monitoring kit can display temperature but leaves the operator to decide what temperature and time to use; the module closes that gap.

## 8.3 Business or sustainability model

Three models are plausible. In a **hardware-plus-support** model, the facility purchases a control kit (probes, controller, display) and pays an annual support fee. In a **service** model, a cooperative owns kits and rents them to members per batch. In a **platform** model, the dashboard and traceability features are offered by subscription to groups of facilities and buyers. Because the principal benefit to the processor is fuel and yield saving, the payback period is the key commercial parameter. It can be estimated for a facility as kit cost divided by annual saving, where annual saving = batches per year × (fuel saving per batch + value of retained yield per batch).

**Table 8.3: Illustrative payback framework (values to be supplied by the facility).**

| Quantity | Source | Example from this study |
|---|---|---|
| Energy saving per batch | 3.82 − 2.94 MJ/kg × 15 kg | 13.2 MJ |
| Yield retained per batch | (6.8% − 1.1%) × 15 kg | approximately 0.85 kg |
| Fuel price | Facility records | [enter local price] |
| Product price | Market records | [enter local price] |
| Kit cost | Quotation | [enter] |
| Batches per year | Facility schedule | [enter] |

### 8.3.1 Go-to-market approach

A staged approach is recommended. In the first stage, pilot installations at a small number of facilities (including the trial facility) are used to collect further batch data, refine the model and gather operator feedback. In the second stage, a standard kit and training package is offered through fisheries cooperatives and local agencies. In the third stage, the platform features (dashboard, traceability, buyer-facing records) are offered on subscription. At each stage, the key performance indicators are energy per kg, loss percentage, boiling-time variability and operator adoption.

### 8.3.2 Commercial risks

**Table 8.4: Commercial risks and mitigations.**

| Risk | Mitigation |
|---|---|
| Low willingness to pay in a low-margin sector | Emphasise payback from fuel and yield; use leasing or cooperative ownership |
| Operator resistance to change | Training; simple interface; demonstrate benefit on the operator's own batches |
| Model invalid for other species or vats | Offer recalibration service; document the valid range |
| Hardware damage in harsh environment | Robust enclosures; spare probes; maintenance plan |
| Regulatory change | Keep records compatible with hazard-analysis documentation |

## 8.4 Scalability, intellectual property and compliance

**Scalability.** The module scales by replication: each vat needs its own probes and controller, while the model and dashboard are shared. Recalibration is required for new species, cut geometries or vat designs, but is inexpensive because the model has few parameters. **Intellectual property.** The empirical constants, the freshness-adaptive logic and the batch data schema are the principal assets. Protection may take the form of copyright in the software, trade-secret treatment of calibrated constants and, where appropriate, patent advice on the combined method. This must be confirmed with the institute's technology-transfer office. **Compliance.** The product must comply with applicable food-safety regulations and electrical safety requirements for equipment operating near steam and water, and the facility remains responsible for hazard analysis. Data protection obligations are minimal because no personal data are collected.

## 8.5 Contribution to the Sustainable Development Goals

**Table 8.5: Contribution to the United Nations Sustainable Development Goals [19].**

| SDG | Contribution |
|---|---|
| SDG 2 – Zero Hunger | Reduces processing loss (flaking 6.8% → 1.1%) and improves the stability of a preserved protein food |
| SDG 7 – Affordable and Clean Energy / SDG 13 – Climate Action | Lowers thermal energy per kg by 23.0%, reducing fuel use and associated emissions |
| SDG 8 – Decent Work and Economic Growth | Raises productivity and product consistency in a traditional livelihood sector |
| SDG 9 – Industry, Innovation and Infrastructure | Brings low-cost digital technology to small processors |
| SDG 12 – Responsible Consumption and Production | Reduces waste and enables traceable production |
| SDG 14 – Life Below Water | Higher yield per fish landed reduces pressure to catch more for the same output |

# 9 CONCLUSION AND FUTURE DIRECTIONS

## 9.1 Conclusion

This thesis presented the design, implementation and evaluation of an intelligent boiling and brine-salting optimisation module for Maldive fish processing, developed as the second tier of an integrated AI and IoT-based smart processing system. The module computes boiling duration from an empirical regression of thickness, batch mass and freshness grade, derives the salt mass for a 12%–15% salt-to-water brine, and supervises core temperature between 92 °C and 96 °C with digital PT100 probes.

Evaluated on 20 production batches of 15 kg against conventional manual practice, the module reduced mean boiling time from 58.4 to 42.1 min (−27.9%), reduced energy consumption from 3.82 to 2.94 MJ/kg (−23.0%), lowered the coefficient of variation of deep-tissue salt penetration from 18.6% to 5.2%, raised core muscle firmness from 3.4 to 4.6, and cut flaking and over-boil loss from 6.8% to 1.1%. These results demonstrate that a transparent, low-cost model can substantially improve a stage that was previously governed by subjective experience.

## 9.2 Achievement of objectives

**Table 9.1: Achievement of the sub-objectives.**

| Sub-objective | Achievement |
|---|---|
| 1. Characterise the stage and variables | Achieved (Chapters 1–3) |
| 2. Empirical boiling-time model | Achieved: Equation (4.1), verified in Table 5.3 |
| 3. Brine-salting rule (12%–15%) | Achieved: Equation (4.2); CV 18.6% → 5.2% |
| 4. PT100 temperature supervision (92–96 °C) | Achieved (Section 4.4.3, Chapter 5) |
| 5. Integration with intake tier and dashboard | Achieved (Section 5.3) |
| 6. Evaluation on 20 batches | Achieved (Chapter 6) |
| 7. Sensitivity, limitations and commercial analysis | Achieved (Sections 4.4.1, 7.5, Chapter 8) |

## 9.3 Individual research and product contributions

The research contribution is the empirical, explainable model and brine rule, together with field evidence of their benefit and the demonstration that upstream AI grading can drive downstream process parameters. The product contribution is the deployable

control and logging module whose batch records connect the intake and drying tiers and enable end-to-end digital traceability in artisanal food processing.

### 9.3.1 Lessons learned

1. Starting from the process objectives and measurable outcomes (time, energy, salt uniformity, firmness, loss) kept the work focused.
2. A simple, transparent model was sufficient to capture the main effects and was easy to explain to operators.
3. Combining a feed-forward model with a measured feedback constraint made the module robust to errors in inputs.
4. Early agreement on interfaces with the other tiers avoided integration problems.
5. Reporting variability as well as means revealed benefits that averages alone would have hidden.

## 9.4 Future directions

1. **Model refinement:** report confidence intervals for the constants; fit the exponents on a larger multi-facility dataset; test additional predictors such as initial fish temperature and fat content.
2. **Closed-loop actuation:** connect the supervision loop to burner or heater actuators so that temperature is held in the 92–96 °C window automatically.
3. **Salt profiling:** integrate multispectral near-infrared (NIR) sensors for non-destructive core salt profiling, as proposed for the wider project [17].
4. **Mechanistic hybrid models:** combine the empirical model with a simplified diffusion model (Equation 2.1) to improve extrapolation.
5. **Controlled study of drying impact:** vary only the boiling method and measure drying time and shelf-life outcomes to quantify the downstream benefit.
6. **Operator study:** run a structured usability and adoption study with operators and owners.
7. **Robotic handling:** automate loading and unloading, complementing the proposed robotic sorting at intake.

8. **Broader validation:** test other species, vat designs and seasons, and run formal statistical tests on per-batch data.

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# GLOSSARY

**Table 9.2: Glossary of terms.**

| Term | Definition |
|---|---|
| **Arrhenius model** | Equation describing how reaction rate depends on temperature; used in the group's shelf-life model. |
| **Batch record** | Digital record of the parameters, temperatures, energy and outcomes of one processed batch. |
| **Boiling duration ($t_b$)** | Time for which fish are held in the boiling brine, in minutes. |
| **Brine** | Solution of salt (NaCl) in water used for boiling and salting. |
| **Case-hardening** | Formation of a dry, hard outer layer that traps moisture inside the product. |
| **Coefficient of variation (CV)** | Standard deviation divided by the mean, expressed as a percentage. |
| **Deep-tissue salt** | Sodium chloride content in the inner (core) muscle of the fish. |
| **DHT22** | Digital temperature and relative humidity sensor. |
| **ESP32** | Low-power microcontroller with Wi-Fi used in the IoT edge nodes. |
| **Flaking loss** | Loss of muscle material through fibre separation caused by over-boiling or agitation. |
| **Freshness grade** | Raw-intake classification: Very Fresh, Fresh or Spoiled. |
| **Late fusion** | Combination of decisions from several models or sensors after each has produced its own output. |
| **MQTT** | Lightweight publish/subscribe messaging protocol for IoT. |
| **NaCl** | Sodium chloride, common salt. |
| **Organoleptic evaluation** | Assessment of a product by human senses. |
| **PT100** | Platinum resistance temperature sensor with 100 Ω resistance at 0 °C. |
| **Random Forest (RF)** | Machine learning ensemble method used for finished product quality evaluation and process optimization. |
| **Salt penetration** | Migration of salt from the brine into the fish muscle. |
| **Thermal offset (δ)** | Extra boiling time added for Fresh batches to ensure core sterilisation. |
| **Water activity ($a_w$)** | Ratio of the vapour pressure of water in a product to that of pure water. |

# APPENDICES

## Appendix A Research instruments or extended results

This appendix provides extended model values, the parameter table and a batch record template.

**Table A.1: Model constants and parameters.**

| Symbol | Meaning | Value |
|---|---|---|
| $\alpha$ | Base boiling time at reference conditions | 40.0 min |
| $d_0$ | Reference mean thickness | 4.5 cm |
| $M_0$ | Reference batch mass | 15 kg |
| Thickness exponent | Exponent on $d_m/d_0$ | 1.4 |
| Mass exponent | Exponent on $M_{batch}/M_0$ | 0.3 |
| $\delta$ (Very Fresh) | Thermal offset | 0 min |
| $\delta$ (Fresh) | Thermal offset | +3.5 min |
| Brine window | Salt-to-water weight ratio | 12%–15% |
| Deep-tissue salt target | NaCl | ≥ 12% |
| Core temperature window | PT100 core temperature | 92–96 °C |

**Table A.2: Extended boiling-duration table, Very Fresh ($\delta$ = 0), minutes. Add 3.5 min for Fresh.**

| $d_m$ (cm) | 5 kg | 10 kg | 15 kg | 20 kg | 25 kg | 30 kg |
|---|---|---|---|---|---|---|
| 3.0 | 16.3 | 20.1 | 22.7 | 24.7 | 26.4 | 27.9 |
| 3.5 | 20.2 | 24.9 | 28.1 | 30.7 | 32.8 | 34.6 |
| 4.0 | 24.4 | 30.0 | 33.9 | 37.0 | 39.5 | 41.8 |
| 4.5 | 28.8 | 35.4 | 40.0 | 43.6 | 46.6 | 49.2 |
| 5.0 | 33.3 | 41.0 | 46.4 | 50.5 | 54.0 | 57.1 |
| 5.5 | 38.1 | 46.9 | 53.0 | 57.7 | 61.7 | 65.2 |
| 6.0 | 43.0 | 53.0 | 59.8 | 65.2 | 69.7 | 73.7 |

**Table A.3: Salt mass (kg) for common water masses.**

| Water (kg) | 12% | 13.5% | 15% |
|---|---|---|---|
| 20 | 2.4 | 2.70 | 3.0 |
| 40 | 4.8 | 5.40 | 6.0 |
| 60 | 7.2 | 8.10 | 9.0 |
| 80 | 9.6 | 10.80 | 12.0 |
| 100 | 12.0 | 13.50 | 15.0 |
| 120 | 14.4 | 16.20 | 18.0 |

**Table A.4: Batch record template.**

| Field | Value |
|---|---|
| Batch ID | |
| Date / start–end time | |
| Mass (kg) / thickness (cm) / grade | |
| Computed $t_b$ (min) / $M_s$ (kg) / ratio | |
| Water mass (kg) | |
| Temperature log reference | |
| Energy (MJ/kg) | |
| Salt CV (%) / firmness (1–5) / loss (%) | |
| Operator override (Y/N, reason) | |

**Table A.5: Proposed MQTT message for the batch record (illustrative schema).**

| Key | Example value | Note |
|---|---|---|
| batch_id | B-2026-0001 | Shared across tiers |
| grade | Fresh | From raw-intake tier |
| mass_kg | 15.0 | Weighed at intake |
| thickness_cm | 4.5 | Measured |
| tb_min | 43.5 | Equation (4.1) |
| salt_kg / ratio | 12.0 / 0.12 | Equation (4.2) |
| t_start / t_end | ISO 8601 timestamps | Boiling start and end |
| override | false | True if operator changed the plan |

## Appendix A.1 Operator feedback questionnaire (proposed instrument)

The following instrument is proposed for the operator study recommended in Section 9.4. Respondents rate each statement from 1 (strongly disagree) to 5 (strongly agree).

1. The recommended boiling time was easy to understand.
2. I trusted the recommended boiling time.
3. The salt amount shown was easy to prepare.
4. The temperature alerts were helpful.
5. The module made my work easier compared with my usual method.
6. The module's recommendations matched my own judgement of when the fish was ready.
7. I would continue to use the module in daily production.

8. I would recommend the module to other processors.

Open questions: What was most difficult to use? What would you change? How much time did you spend measuring thickness per batch?

## Appendix B Academic integrity and AI use disclosure

In accordance with the module policy, the use of generative AI is disclosed as follows. [Confirm and edit before submission.] A generative AI assistant (Claude, by Anthropic) was used to help structure and draft the text of this thesis, to generate the figures from the equations and values given in the group research paper, and to compute the derived values and tables from Equation (4.1). All experimental values (Table 6.1) are taken from the group research paper [17]. The candidate is responsible for checking all text, figures, calculations and citations, and for the accuracy, originality and conclusions of the thesis.

Items marked with highlighted square brackets in this document are placeholders for information that is not stated in the group paper and must be completed by the candidate.